

A non-oscillatory time integration method for numerical simulation of stress wave propagations

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Abstract

When the compressive loads are dominant in a composite structure, a tensile stress may be induced owing to the propagation of a stress wave and the interaction between an incident wave and a reflection wave, thus leading to the occurrence of cracks. Therefore, stress wave have a significant effect on the life of composite structures. In this study, a four sub-step explicit time integration scheme is proposed for solving stress wave propagation problems. This method builds on the fourth-order central difference method and a high-order derivative term to minimize the numerical oscillation. The proposed scheme possesses a first-order accuracy in the case of undamped and damped systems. Stability, accuracy, and dispersion of

the proposed explicit direct time integration scheme are analyzed. Furthermore, the performance of this scheme is illustrated by the solution of a stress wave propagation and wave reflection in a one-dimensional impact problem and two-dimensional scalar wave propagation.

Keywords: Stress wave propagation, Finite elements, Explicit time integration, Stability condition, Numerical dispersion, Numerical oscillation

1. Introduction

The propagation of stress waves and wave reflection are important factors in a composite structure system. If a compressive wave propagates to a corner enclosed by two free surfaces at a certain angle, then net tensile stress may occur due to the interaction between the two reflective waves and thus cause the corner spalling [1]. The compressive loads are applied to the composite structure and then transverse cracks caused by the tensile stress may be initiated. Therefore, in the present study, we start the theoretical investigation of the stress wave in order to analysis the crack of structures under the compressive loads.

Stress wave propagation is investigated considering the inertia of the infinitesimal element of a structure under explosive or impact loading. The hyperbolic wave equations are solved by mathematical analysis. As mathematical analysis is performed under limiting conditions, numerical methods can be used for approximate solving of the wave equations. These numerical methods can be categorized into the following three groups [1]: characteristics method, finite difference method, and finite element method.

The characteristics method is usually applied to solve one-dimensional stress wave problems. However, in general multidimensional stress wave problems, as the characteristics and compatibility equations are fairly complicated, it is difficult to obtain a solution. The finite difference method has been widely applied to solve the problems of rectangular or simple shapes, where it is relatively easy to obtain a solution [2]. The feature of the finite element method is that it handles complicated geometries and boundary conditions. This has resulted in extensive finite element research and numerous studies about wave propagation have been performed using this method [2-9].

There are various general sources of errors in the finite element solution [2], such as the discretization [3,4,10-12], integration in space [4-6,13], constitutive relations [14-18], dynamic equilibrium equations [7,8,19-22], iteration [23-26], and round-off [27-29]. In the case of dynamic equilibrium equations, approximate finite element solutions are obtained by direct time integration schemes. Direct time integration is mainly used in stress wave propagation and dynamic structure problems. It is difficult to find accurate finite element solutions for stress wave propagation problems using the direct time integration method. The finite element solutions are not accurate owing to numerical errors occurring in temporal and spatial discretization, such as artificial period elongations, amplitude decays, numerical dispersions, dissipations, and oscillations [2-8].

In previous research, many direct time integration schemes were proposed to improve the accuracy of the solution. The time integral schemes were presented in the central difference method, Runge-Kutta method, Bathe method [7], Noh-Bathe method [8], Newmark method [19], Hulbert and Chung [21], and Tchamwa-Wielgosz method [22]. The Tchamwa-Wielgosz method is first-order accurate, and the central difference method, Bathe method, Noh-Bathe method, and Newmark method are second-order accurate. However, even when these methods are applied to solve the problems, the numerical

oscillations and errors remain. In addition, the four sub-step Runge-Kutta scheme with fourth-order accuracy cause spurious oscillations of the solution [30].

Owing to numerical oscillations, the finite element solutions differ from exact solutions. Therefore, it is difficult to obtain accurate solutions and analyze the results by the finite element analysis of structures. For example, in Ref. [31], the numerical oscillations in contact pressure were presented using the finite element contact model. Furthermore, in Ref. [32], resulting from dynamic response and numerical effect, some numerical oscillations in the impact force appeared. Similarly, de Matos and Nowell [33] reported some difficulties using stress extrapolation, which was used to investigate the plasticity-induced fatigue crack closure according to the corner point singularities and the Poisson's ratio. In Ref. [34], the spurious oscillations may result in low accuracy when estimating a structure life by the Rainflow method. An important point for the fatigue limit state of structures is to reduce the errors from numerical oscillations. It is also important to reduce errors in wave propagation caused by spurious oscillations.

Despite numerous studies using the direct time integration method, there is no reliable integration method for the accurate non-oscillatory solution, even for wave propagation problems in elastic materials, which undermines the reliability of the numerical results in wave propagation solutions. However, in previous research, the finite element problems have been solved by using the filtering processing technique instead of an alternative direct time integration method. By filtering the specific time and space points, the errors from numerical oscillations can be reduced [5,35-36]. However, these approaches do not lend themselves to analyses requiring a solution for all times and over the complete solution domain [7].

In the present study, we propose the explicit time integration scheme to reduce the numerical oscillations in the propagation of stress waves. The procedure for the new explicit time integration scheme with first-order accuracy is developed and evaluated through stability and accuracy analyses, according to standard procedures described in details in Ref. [2]. Then we investigate the dispersion properties in one-dimensional and two-dimensional analyses. Finally, we provide the calculated response for the elastic bar wave problems and a two-dimensional wave problem using the proposed method, Bathe method, and central difference method.

2. Four sub-step explicit time integration scheme

2.1 Proposed explicit time integration scheme

The governing finite element equations in linear analysis are provided by Eq. (1):

$$\mathbf{M}\ddot{\mathbf{U}} + \mathbf{C}\dot{\mathbf{U}} + \mathbf{K}\mathbf{U} = \mathbf{R} \quad (1)$$

where $\mathbf{U}$, $\mathbf{M}$, $\mathbf{C}$, $\mathbf{K}$, and $\mathbf{R}$ are the nodal displacement vector, mass matrix, damping matrix, stiffness matrix and external nodal force vector, respectively, and the over-dot means differential with respect to time. With the displacements, velocities, and accelerations at time t , the solutions at time $t + \Delta t$ are calculated by the direct time integration scheme.

The basic approach of the proposed method is to use the four-order central difference method and a high-order derivative term, as inspired by the Kawamura-Kuwahara scheme [37]. The Kawamura-Kuwahara scheme is a numerical scheme that considers a four-order central difference equation and a four-order

derivative term. In the proposed method, the time step Δt is divided into four sub-steps; then, unknown displacements, velocities, and accelerations are calculated.

In the proposed method, the first sub-step is as follows (see Eq. (2)-(4)):

$${}^{t+\Delta t/4}\mathbf{U} = {}^t\mathbf{U} + \frac{\Delta t}{4} {}^t\dot{\mathbf{U}} + \frac{1}{2} \left(\frac{\Delta t}{4} \right)^2 {}^t\ddot{\mathbf{U}} \quad (2)$$

$${}^{t+\Delta t/4}\dot{\mathbf{U}} = {}^t\dot{\mathbf{U}} + \frac{\Delta t}{4} {}^t\ddot{\mathbf{U}} \quad (3)$$

$$\mathbf{M}^{t+\Delta t/4}\ddot{\mathbf{U}} + \mathbf{C}^{t+\Delta t/4}\dot{\mathbf{U}} + \mathbf{K}^{t+\Delta t/4}\mathbf{U} = {}^{t+\Delta t/4}\hat{\mathbf{R}} \quad (4)$$

The second sub-step is as follows (see Eq. (5)-(7)):

$${}^{t+\Delta t/2}\mathbf{U} = {}^t\mathbf{U} + \frac{\Delta t}{2} {}^{t+\Delta t/4}\dot{\mathbf{U}} + \frac{1}{2} \left(\frac{\Delta t}{2} \right)^2 {}^{t+\Delta t/4}\ddot{\mathbf{U}} \quad (5)$$

$${}^{t+\Delta t/2}\dot{\mathbf{U}} = {}^t\dot{\mathbf{U}} + \frac{\Delta t}{2} {}^{t+\Delta t/4}\ddot{\mathbf{U}} \quad (6)$$

$$\mathbf{M}^{t+\Delta t/2}\ddot{\mathbf{U}} + \mathbf{C}^{t+\Delta t/2}\dot{\mathbf{U}} + \mathbf{K}^{t+\Delta t/2}\mathbf{U} = {}^{t+\Delta t/2}\hat{\mathbf{R}} \quad (7)$$

The third sub-step is as follows (see Eq. (8)-(10)):

$${}^{t+3\Delta t/4}\mathbf{U} = {}^{t+\Delta t/4}\mathbf{U} + \frac{\Delta t}{2} {}^{t+\Delta t/2}\dot{\mathbf{U}} + \frac{1}{2} \left(\frac{\Delta t}{2} \right)^2 {}^{t+\Delta t/2}\ddot{\mathbf{U}} \quad (8)$$

$${}^{t+3\Delta t/4}\dot{\mathbf{U}} = {}^{t+\Delta t/4}\dot{\mathbf{U}} + \frac{\Delta t}{2} {}^{t+\Delta t/2}\ddot{\mathbf{U}} \quad (9)$$

$$\mathbf{M}^{t+3\Delta t/4}\ddot{\mathbf{U}} + \mathbf{C}^{t+3\Delta t/4}\dot{\mathbf{U}} + \mathbf{K}^{t+3\Delta t/4}\mathbf{U} = {}^{t+3\Delta t/4}\hat{\mathbf{R}} \quad (10)$$

Finally, the fourth sub-step is as follows (see Eq. (11)-(13)):

$${}^{t+\Delta t}\mathbf{U} = 8{}^{t+3\Delta t/4}\mathbf{U} - 8{}^{t+\Delta t/4}\mathbf{U} + {}^t\mathbf{U} - 3\Delta t{}^{t+\Delta t/2}\dot{\mathbf{U}} + \alpha(\Delta t)^4 {}^{t+\Delta t/2}\mathbf{U}^{(4)} \quad (11)$$

$${}^{t+\Delta t}\dot{\mathbf{U}} = 8{}^{t+3\Delta t/4}\dot{\mathbf{U}} - 8{}^{t+\Delta t/4}\dot{\mathbf{U}} + {}^t\dot{\mathbf{U}} - 3\Delta t{}^{t+\Delta t/2}\ddot{\mathbf{U}} + \alpha(\Delta t)^4 {}^{t+\Delta t/2}\mathbf{U}^{(5)} \quad (12)$$

$$\mathbf{M}{}^{t+\Delta t}\ddot{\mathbf{U}} + \mathbf{C}{}^{t+\Delta t}\dot{\mathbf{U}} + \mathbf{K}{}^{t+\Delta t}\mathbf{U} = {}^{t+\Delta t}\widehat{\mathbf{R}} \quad (13)$$

where $\mathbf{U}^{(i)}$ is i -order differential and α is a parameter to be determined. The parameter α affects the stability limit and accuracy of the solution. ${}^{t+\Delta t/4}\widehat{\mathbf{R}}$, ${}^{t+\Delta t/2}\widehat{\mathbf{R}}$, and ${}^{t+3\Delta t/4}\widehat{\mathbf{R}}$ are the load vectors corresponding to each sub-step and are given in Section 2.5. ${}^{t+\Delta t/2}\mathbf{U}^{(4)}$ and ${}^{t+\Delta t/2}\mathbf{U}^{(5)}$ are calculated using the following equations (see Eq. (14)-(15)):

$${}^{t+\Delta t/2}\mathbf{U}^{(4)} = \frac{{}^{t+3\Delta t/4}\ddot{\mathbf{U}} - 3{}^{t+\Delta t/2}\ddot{\mathbf{U}} + 3{}^{t+\Delta t/4}\ddot{\mathbf{U}} - {}^t\ddot{\mathbf{U}}}{(\Delta t)^3/32} \quad (14)$$

$${}^{t+\Delta t/2}\mathbf{U}^{(5)} = \frac{{}^{t+3\Delta t/4}\dddot{\mathbf{U}} - 3{}^{t+\Delta t/2}\dddot{\mathbf{U}} + 3{}^{t+\Delta t/4}\dddot{\mathbf{U}} - {}^t\dddot{\mathbf{U}}}{(\Delta t)^3/32} \quad (15)$$

The first sub-step can be seen as the explicit Euler forward method. The second and third sub-steps are based on the second-order central difference method, while the fourth sub-step can be interpreted as using the fourth-order central difference method and a high-order derivative term. The fourth-order and fifth-order differentials are derived from the second-order central difference method and the Euler backward method.

2.2 Stability analysis

In the modal approach, we decompose the displacement response by using Eq. (16).

$$\mathbf{U}(t) = \boldsymbol{\Psi}\mathbf{X}(t) \quad (16)$$

where Ψ is the modal matrix of the free-vibration modes and $\mathbf{X}(t)$ is the generalized modal displacement. The modal matrix Ψ is acquired by the following eigenvalue problem (see Eq. (17)-(18)):

$$\mathbf{K}\Psi = \mathbf{M}\Psi\Lambda \quad (17)$$

$$\Lambda = \text{diag}(\omega_1^2, \omega_2^2, \dots, \omega_n^2) \quad (18)$$

where Λ is a diagonal matrix of the eigenvalues and ω_i is the i -th natural frequency component of Eq. (1). Substituting Eq. (16) into Eq. (1), we obtain the following equation (Eq. (19)):

$$\ddot{x}_i(t) + 2\omega_i\xi_i\dot{x}_i(t) + x_i(t) = r_i(t), \quad i = 1, 2, \dots, n \quad (19)$$

where ξ_i is the i -th damping ratio and r_i is the generalized force ($= \Psi(i,:)^\top \mathbf{R}$). In the modal equations, the recursive relationship of the proposed method is as follows (see Eq. (20)) [2]:

$$\begin{bmatrix} {}^{t+\Delta t}\ddot{\mathbf{x}} \\ {}^{t+\Delta t}\dot{\mathbf{x}} \\ {}^{t+\Delta t}\mathbf{x} \end{bmatrix} = \mathbf{A} \begin{bmatrix} {}^{t+\Delta t}\ddot{\mathbf{x}} \\ {}^{t+\Delta t}\dot{\mathbf{x}} \\ {}^{t+\Delta t}\mathbf{x} \end{bmatrix} + \mathbf{L}_a {}^{t+\Delta t/4}\mathbf{r} + \mathbf{L}_b {}^{t+\Delta t/2}\mathbf{r} + \mathbf{L}_c {}^{t+3\Delta t/4}\mathbf{r} + \mathbf{L}_d {}^{t+\Delta t}\mathbf{r} \quad (20)$$

where matrix $\mathbf{A}$ and vectors $\mathbf{L}$ are the integration approximation and load operators, respectively. The operators $\mathbf{A}$, $\mathbf{L}_a$, $\mathbf{L}_b$, $\mathbf{L}_c$, and $\mathbf{L}_d$ are given in Appendix A. The stability and accuracy analyses are carried out with Eq. (20).

Considering the case where $\xi = 0.0$, the characteristic polynomial of $\mathbf{A}$ becomes as follows (see Eq. (21)):

$$p(\lambda) = \lambda^3 - 2A_1\lambda^2 + A_2\lambda + A_3 \quad (21)$$

where, as shown in Eq. (22)-(24),

$$A_1 = 1 - \omega_0^2(\Delta t)^2 + \frac{19+224\alpha}{128}\omega_0^4(\Delta t)^4 - \frac{1+48\alpha}{512}\omega_0^6(\Delta t)^6 \quad (22)$$

$$A_2 = 1 - \omega_0^2(\Delta t)^2 + \frac{11+992\alpha}{64}\omega_0^4(\Delta t)^4 + \frac{1-968\alpha+4096\alpha^2}{128}\omega_0^6(\Delta t)^6 + \frac{\alpha(640\alpha+9)}{64}\omega_0^8(\Delta t)^8 \quad (23)$$

$$A_3 = 0 \quad (24)$$

and ω_0 is the modal natural frequency. Since $A_3 = 0$, the eigenvalues of $\mathbf{A}$ are as follows (see Eq. (25)):

$$\lambda_{1,2} = A_1 \pm \sqrt{A_1^2 - A_2} \text{ and } \lambda_3 = 0 \quad (25)$$

For stability, we need that the spectral radius of the approximation operator be smaller than or equal to 1 [2]. The spectral radius of $\mathbf{A}$ is defined as follows (see Eq. (26)):

$$\rho(\mathbf{A}) = \max(|\lambda_1|, |\lambda_2|, |\lambda_3|) \quad (26)$$

Mathematically, it is difficult to solve the eigenvalues of $\mathbf{A}$ in Eq. (25), because $A_1^2 - A_2$ is the sextic equation about $\omega_0^2(\Delta t)^2$. As an alternative method for finding the eigenvalues, a numerical algorithm can be considered [38]. In this study, the spectral radius $\rho(\mathbf{A})$ about $\Omega_0 (= \omega_0\Delta t)$ for various α is calculated using the regula-falsi method, one of the numerical methods. When the spectral radius of $\mathbf{A}$

is equal to 1, Ω_0 is called the stability limit and is given by $\Omega_s = \omega_0 \Delta t_{cr}$, where Δt_{cr} is the critical time step. As shown in Fig. 1, the parameter α plays an important role in stabilization behavior. The maximum stability limit is ca. 2.259 with the corresponding $\alpha \approx 0.045$. If high-order derivative terms in Eq. (14)-(15) are not considered (i.e. $\alpha = 0$), the stability limit is ca. 1.384, which is lower than the stability limit of the central difference method. Thus, the proposed method can have a high stability limit using the appropriate parameter α (for stability, we recommend using $\alpha = 0.045$). The important point in the proposed method is that the spectral radius of the approximation operator $\rho(\mathbf{A})$ rapidly decreases with numerical damping after Ω_0 is ca. 0.15.

There is a non-differentiable point when Ω_0 is the bifurcation point (see Fig. 2). The bifurcation point is the specific point where the eigenvalues change from real to complex or vice versa, i.e. $A_1^2 - A_2 = 0$. The bifurcation point Ω_b for various α acquired by the regula-falsi method is given in Fig. 1, where Ω_{b1} and Ω_{b2} are the first and second bifurcation points, respectively. The spectral radius at the first bifurcation point is denoted by ρ_b . With $\rho_b = 1$, there is no numerical dissipation, while $\rho_b = 0$ results in maximum numerical dissipation [8]. The range of the parameter α for $0 \leq \rho_b \leq 1$ is obtained as $0.0409 \leq \alpha \leq 0.045$ (see Fig. 3).

Fig. 2 shows that ρ_b of the proposed method decreases with a decrease of the bifurcation limit Ω_b . The decrement of the bifurcation limit in the proposed method is considerably smaller than in other methods. When the value of ρ_b changes from 0 to 1 (from maximum to minimum dissipation), the bifurcation limit Ω_b of the proposed method changes from 1.917 to 2.257, while Ω_b of the Noh–Bathe method [8] changes from 3.414 to 4, Ω_b of Hulbert and Chung [21] changes from 1.414 to 2, and Ω_b of the Tchamwa-Wielgosz scheme [22] changes from 1 to 2.

As shown in Fig. 4, the proposed method is highly dissipative even at low frequencies, while the Tchamwa-Wielgosz is low dissipative at low frequencies, the central difference method is a non-dissipative time integration method. The stability limit of the proposed method is larger than that of the Tchamwa-Wielgosz and central difference method. As the proposed method has four sub-steps, the computational effort is approximately four times that of the central difference method. In contrast, the Noh-Bathe method and Bathe method consist of two sub-steps. The Noh-Bathe method has a stability limit of approximately twice that of the central difference method, and the Bathe method is unconditionally stable. Both the Noh-Bathe method and Bathe method exhibit low dissipation at low frequencies.

2.3 Accuracy analysis

We can investigate the accuracy characteristics of the proposed method by solving the following initial value problem (see Eq. (27)) [2]:

$$\begin{cases} \ddot{x} + \omega^2 x = 0 \\ {}^0x = 1.0; \quad {}^0\dot{x} = 0.0; \quad {}^0\ddot{x} = -\omega^2 \end{cases} \quad (27)$$

The exact solution of Eq. (27) is $x = \cos\omega t$. Although the influence of the damping ratio ξ needs to be evaluated, the significant solution characteristics can be investigated by solving the solution of Eq. (27) [2]. Using Eq. (20), the numerical solution of the proposed method can be calculated. Fig. 5 shows the period elongations and amplitude decays for various values of parameter α . When $\alpha = 0.045$, the period elongations are larger and the amplitude decays are smaller than those with $\alpha = 0.0409$.

However, the difference in period elongations and amplitude decays appears to be small when $0.0409 \leq \alpha \leq 0.045$. For general use, we suggest setting $\alpha = 0.045$, which results in a high stability limit and minimum dissipation. The important point to note here is that, until $\Delta t/T$ is ca. 0.12, the amplitude decay increases with an increase of $\Delta t/T$, and thereafter it decreases in the proposed method. For the proposed method, larger solution errors in the period elongation and amplitude decay are observed as compared with those of other methods for the same $\Delta t/T$ (see Fig. 6). When the period elongation has an accuracy of ca. 1%, $\Delta t/T$ for the proposed method ($\alpha = 0.045$) is 0.03, while $\Delta t/T$ for the Tchamwa-Wielgosz method ($\varphi = 1.033$) is 0.08, $\Delta t/T$ for the Noh-Bathe method ($p = 0.54$) is 0.19, and $\Delta t/T$ for Bathe method is 0.08. In addition, when $\Delta t/T$ is 0.05 the amplitude decay for the proposed method ($\alpha = 0.045$) is 60%, whereas that in the case of the Noh-Bathe method and Bathe method is negligible.

2.4 Dispersion analysis for the one-dimensional case

In this section, we analyze the dispersion errors using the proposed method, Bathe method, and central difference method for a one-dimensional wave propagation problem. The regular mesh of a 2-node element is used for the spatial discretization.

The scalar wave governing equation is given by Eq. (28):

$$\frac{\partial^2 u}{\partial t^2} - c_0^2 \nabla^2 u = 0 \quad (28)$$

where u is the field variable, c_0 is the exact wave velocity, and ∇^2 is the Laplace operator. The corresponding finite element discretization gives Eq. (29):

$$\mathbf{M}\ddot{\mathbf{U}} + c_0^2 \mathbf{K}\mathbf{U} = \mathbf{0} \quad (29)$$

where $\mathbf{M}$ and $\mathbf{K}$ are the global mass and stiffness matrices. The matrices $\mathbf{M}$ and $\mathbf{K}$ can be acquired by the summation process using the matrices $\mathbf{M}^{(m)}$ and $\mathbf{K}^{(m)}$ [2]. The matrices $\mathbf{M}^{(m)}$ and $\mathbf{K}^{(m)}$ are the local mass and stiffness matrices of element (m) , respectively, with the volume $V^{(m)}$ as in Eq. (30)-(31).

$$\mathbf{M}^{(m)} = \int_{V^{(m)}} \rho^{(m)} \mathbf{N}^{(m)T} \mathbf{N}^{(m)} dV \quad (30)$$

$$\mathbf{K}^{(m)} = \int_{V^{(m)}} \mathbf{B}^{(m)T} \mathbf{C}^{(m)} \mathbf{B}^{(m)} dV \quad (31)$$

where $\rho^{(m)}$, $\mathbf{N}^{(m)}$, $\mathbf{B}^{(m)}$, and $\mathbf{C}^{(m)}$ are the density, displacement shape (or interpolation) matrix, strain-displacement matrix, and elasticity matrix of element (m) , respectively. In the one-dimensional elastic problem case, the elasticity matrix $\mathbf{C}^{(m)}$ is simply the Young's modulus E . For actual evaluation, we use the unit density and the unit cross-sectional area and assume that $E = 1$ in the one-dimensional solution.

Using Eq. (2)-(15), the linear multistep form of the proposed explicit scheme is expressed as follows (see Eq. (32)-(33)):

$$\mathbf{M}^{t+\Delta t} \mathbf{U} - 8\mathbf{M}^{t+3\Delta t/4} \mathbf{U} - 16\alpha c_0^2 (\Delta t)^2 \mathbf{K}^{t+\Delta t/2} \mathbf{U} + \left[20\mathbf{M} + \left(\frac{3}{2} + 48\alpha \right) c_0^2 (\Delta t)^2 \mathbf{K} \right]^{t+\Delta t/4} \mathbf{U}$$

$$-\left[13\mathbf{M} + \left(\frac{3}{8} + 32\alpha\right) c_0^2(\Delta t)^2 \mathbf{K}\right] {}^t\mathbf{U} = \mathbf{0} \quad (32)$$

or

$$\begin{aligned} & \mathbf{M}^{t+\Delta t} \mathbf{U} - 8\mathbf{M}^{t+3\Delta t/4} \mathbf{U} - 16\alpha \gamma \mathbf{K}^{t+\Delta t/2} \mathbf{U} + \left[20\mathbf{M} + \left(\frac{3}{2} + 48\alpha\right) \gamma \mathbf{K}\right] {}^{t+\Delta t/4} \mathbf{U} \\ & - \left[13\mathbf{M} + \left(\frac{3}{8} + 32\alpha\right) \gamma \mathbf{K}\right] {}^t\mathbf{U} = \mathbf{0} \end{aligned} \quad (33)$$

where $\gamma = \text{CFL}^2 h^2$, CFL is the CFL number $\left(= \frac{c_0 \Delta t}{h}\right)$, and h is the characteristic length of the finite element. In the one-dimensional solution, the characteristic length h is the length of element Δx .

In the Bathe method, the linear multistep form is given by the following equations (see Eq. (34)-(35)) [7]:

$$\begin{aligned} & (72\mathbf{M} + 8c_0^2(\Delta t)^2 \mathbf{K}) {}^{t+\Delta t} \mathbf{U} + (-144\mathbf{M} + 5\alpha c_0^2(\Delta t)^2 \mathbf{K}) {}^{t+\Delta t/2} \dot{\mathbf{U}} \\ & + (72\mathbf{M} + 5c_0^2(\Delta t)^2 \mathbf{K}) {}^t \mathbf{U} = \mathbf{0} \end{aligned} \quad (34)$$

or

$$(72\mathbf{M} + 8\gamma \mathbf{K}) {}^{t+\Delta t} \mathbf{U} + (-144\mathbf{M} + 5\gamma \mathbf{K}) {}^{t+\Delta t/2} \mathbf{U} + (72\mathbf{M} + 5\gamma \mathbf{K}) {}^t \mathbf{U} = \mathbf{0} \quad (35)$$

In the central difference method, the linear multistep form is expressed by the following (see Eq. (36)-(37)) [2]:

$$\mathbf{M}^{t+\Delta t} \mathbf{U} + [-2\mathbf{M} + c_0^2(\Delta t)^2 \mathbf{K}] {}^t \mathbf{U} + \mathbf{M}^{t-\Delta t} \mathbf{U} = \mathbf{0} \quad (36)$$

or

$$\mathbf{M}^{t+\Delta t} \mathbf{U} + [-2\mathbf{M} + \gamma \mathbf{K}] {}^t \mathbf{U} + \mathbf{M}^{t-\Delta t} \mathbf{U} = \mathbf{0} \quad (37)$$

In the one-dimensional analysis, the general solution of Eq. (28) along the x coordinate is $u = Ae^{i(kx-\omega t)}$, where ω , $k = \omega/c$, and c are the numerical frequency, wave number, and numerical wave speed, respectively. The numerical wave speed c is not the same as the exact wave speed c_0 . The numerical dispersion is observed by the differences in wave speeds.

Considering the uniform mesh with 2-node equally spaced Δx along the x axis, the solution of the wave equation at time $n_t \Delta t$ and space $n_x \Delta x$ is obtained by the following equation (see Eq. (38)) [7]:

$$\frac{n_t \Delta t}{n_x \Delta x} u = A_k e^{i(kn_x \Delta x - \omega n_t \Delta t)} = A_k e^{ik \Delta x (n_x - n_t (CFL)(c/c_0))} \quad (38)$$

For a 2-node element, the consistent mass matrix $\mathbf{M}_c$, lumped mass matrix $\mathbf{M}_l$, and stiffness matrix $\mathbf{K}$ are the symmetric matrices given by Eq. (39)-(41).

$$\mathbf{M}_c = \frac{\Delta x}{6} \begin{pmatrix} 2 & 1 & 0 & 0 & \cdots & 0 \\ & 4 & 1 & 0 & \cdots & 0 \\ & & 4 & 1 & & \vdots \\ & & & \ddots & & \vdots \\ & \text{syms} & & & 4 & 1 \\ & & & & & 2 \end{pmatrix} \quad (39)$$

$$\mathbf{M}_l = \frac{\Delta x}{2} \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ & 2 & 0 & 0 & \cdots & 0 \\ & & 2 & 0 & & \vdots \\ & & & \ddots & & \vdots \\ & \text{syms} & & & 2 & 0 \\ & & & & & 1 \end{pmatrix} \quad (40)$$

$$\mathbf{K} = \frac{1}{\Delta x} \begin{pmatrix} 1 & -1 & 0 & 0 & \cdots & 0 \\ & 2 & -1 & 0 & \cdots & 0 \\ & & 2 & -1 & & \vdots \\ & & & \ddots & & \vdots \\ & & & & 2 & -1 \\ & & & & & 1 \end{pmatrix} \quad (41)$$

In the explicit and implicit time integration schemes, the constant and lumped mass matrices affect the finite element solutions and the dispersion errors. Ham and Bathe [6] proposed that the mass matrix be changed slightly, in order to obtain accurate solutions. As mentioned in Ref. [39], the use of a lumped mass matrix or a combination of the lumped and consistent mass matrices might cause a reduction in the errors. Therefore, we study the dispersion properties of the lumped and consistent mass matrices.

2.4.1. Consistent mass matrix

Substituting Eq. (39) and (41) by Eq. (33), (35), and (37), the relation of the CFL number, ratio of numerical wave speed and exact wave speed c/c_0 , wave number k , and element length Δx is obtained for the proposed method, Bathe method, and central difference method using the consistent mass matrix. Fig. 7 shows the relative wave speed errors of the proposed method ($\alpha = 0.045$). When the CFL number is 0.22, all wave modes with $k\Delta x > 0.7$ have dispersion, while all lower frequency modes have non-dispersion. For the Bathe method (CFL=1), the condition that the wave modes have dispersion is $k\Delta x > 2.3$ (see Fig. 8). Compared to the limit of wave modes having non-dispersion, the range of non-dispersive wave modes in the proposed method is smaller than that yielded by the Bathe method. As can be seen in Fig. 9, we obtain the relative wave speed errors in the central difference method for various values of CFL. For the central difference method, the relative wave speed errors are

always positive (see Fig. 9) and the maximum value of the relative wave speed errors increase with an increase in the CFL number for $0 \leq \frac{\Delta x}{\lambda/2} \leq 1$. When the CFL number decreases, a smaller time step size is selected, which may result in high calculation cost. As shown in Fig. 10, the maximum values of the relative wave speed errors for the central difference method are converged after the CFL number is ca. 0.1.

The wave modes are discarded due to numerical damping in the numerical solutions. In the proposed method, high frequency wave modes with $\Omega_0 > 0.15$ result in numerical damping, while wave modes with $\Omega_0 > 0.6\pi$ are discarded for the Bathe method [7]. However, there are no discarded wave modes in the central difference method, as there is no numerical damping. Using the definition of CFL number, $k\Delta x$ in terms of Ω_0 is expressed as follows (see Eq. (42)):

$$k\Delta x = \frac{c_0}{c} \frac{\Omega_0}{\text{CFL}} \quad (42)$$

When $\text{CFL} = 0.22$ for the proposed method and $\text{CFL} = 1$ for the Bathe method, we assume that the numerical wave speed is approximately equal to the exact wave speed ($c_0/c \approx 1$). The discarded wave modes of the proposed method and Bathe method are the wave modes with $k\Delta x > 0.68$ and $k\Delta x > 0.6\pi$ using Eq. (42). As the total solution is mostly non-dispersive, the solutions of the proposed method ($\text{CFL} = 0.22$) and Bathe method ($\text{CFL} = 1$) are very accurate.

2.4.2. Lumped mass matrix

Here, the dispersion analysis for the lumped mass matrix is performed. When Eq. (40) is used instead of Eq. (39), then the dispersion properties are studied by similar processes, as in the consistent mass matrix. Fig. 11-13 show the relative wave speed errors of the proposed method ($\alpha = 0.045$), Bathe method, and central difference method, respectively, for various CFL numbers. Comparing the relative wave speed errors for the consistent mass and lumped mass, the dispersion errors decrease using the consistent mass in the proposed method and Bathe method, while the central difference method with lumped mass have lower dispersion errors than with consistent mass.

Considering the same value of CFL number, the minimum discarded mode of the proposed method is smaller than that of the Bathe method, as the proposed method has the smaller value of Ω_0 , at which the spectral radius begins to rapidly decrease and numerical damping starts. As the relative wave speed errors of the proposed method and Bathe method do not have an approximate value of 0 for the participating (effective) wave modes, the dispersion error in the proposed method may be smaller than in the Bathe method, with the same CFL number.

For the proposed method and the Bathe method, the relative wave speed errors are negative for any CFL number (see Fig. 11-12) and the maximum (negative) value of the relative wave speed errors decreases with an increase in the CFL number. However, when the CFL number increases, a larger time step size is selected, which may result in poor accuracy. As shown in Fig. 14 – 15, the maximum values of the relative wave speed errors for the proposed method and Bathe method are converged after the CFL number is ca. 0.5 and ca. 3, respectively.

For the proposed method with CFL = 0.5, the range of participating wave modes is ca. $k\Delta x < 0.3$ and the maximum value of the relative wave speed errors is ca. 0.02. For the Bathe method with CFL = 3, the maximum speed error is ca. 0.11, and the range of discarded wave modes is ca. $k\Delta x < 0.2\pi$. Considering the lumped mass matrix, the effective wave modes of the proposed method are smaller than those available with the Bathe method; however, the Bathe method has the maximum value of the relative wave speed errors almost fifth that of the proposed method.

2.5 Dispersion analysis in the two-dimensional case

In the two-dimensional wave propagation problem, the general solution and corresponding numerical solution of Eq. (28) are given by the Eq. (43) and Eq. (44), respectively:

$${}_{x,y}^t u = A_k e^{i(k_0 x \cos \theta + k_0 y \sin \theta - \omega_0 t)} \quad (43)$$

$${}_{x,y}^t u = A_k e^{i(kx \cos \theta + ky \sin \theta - \omega t)} \quad (44)$$

Using the regular mesh of a 4-node element with nodes equally spaced at a distance of h from each other along the x and y axes ($\Delta x = \Delta y = h$) for the spatial discretization, the solution of the wave equation at time $n_t \Delta t$ and in space $n_x h, n_y h$ is obtained by using the following equation (see Eq. (45)) [7,8]:

$${}_{n_x h, n_y h}^{n_t \Delta t} u = A_k e^{i(k n_x h \cos \theta + k n_y h \sin \theta - \omega n_t \Delta t)} = A_k e^{ikh(n_x \cos \theta + n_y \sin \theta - n_t (\text{CFL})(c/c_0))} \quad (45)$$

where θ is the propagating angle measured from the x axis.

For the 4-node element, the rows of the consistent mass matrix $\mathbf{M}_c$, lumped mass matrix $\mathbf{M}_l$,

and stiffness matrix $\mathbf{K}$ are obtained as follows (see Eq. (46)-(48)) [7]:

$$\text{Row}(\mathbf{M}_c) = \frac{h^2}{36} [0 \ \cdots \ 0 \ 1 \ 4 \ 1 \ 4 \ 16 \ 4 \ 1 \ 4 \ 1 \ 0 \ \cdots \ 0] \quad (46)$$

$$\text{Row}(\mathbf{M}_l) = h^2 [0 \ \cdots \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ \cdots \ 0] \quad (47)$$

$$\text{Row}(\mathbf{K}) = \frac{1}{3} [0 \ \cdots \ 0 \ -1 \ -1 \ -1 \ -1 \ 8 \ -1 \ -1 \ -1 \ -1 \ 0 \ \cdots \ 0] \quad (48)$$

When Eq. (46)-(48) are used instead of Eq. (39)-(41), the dispersion analysis for the two-dimensional case is studied using a similar process, as that of the one-dimensional case. In order to study the influence of the mass matrix on the two-dimensional dispersion properties, both the consistent mass and lumped mass matrices are considered.

In the dispersion analysis of the consistent mass matrix, the considered CFL numbers for the proposed method ($\alpha = 0.045$) and Bathe method are 0.22 and 0.1, respectively, which results in almost non-dispersive solutions. For the central difference method, we consider $\text{CFL} = 1$, which results in no dispersion error. Fig. 16-18 show the dispersion error of the proposed method ($\alpha = 0.045$), Bathe method, and central difference method, respectively, for propagation angle. For the proposed method and other methods, lower curves are shown in Fig. 16-18 with the increase in the angles. However, the proposed method shows minimal variation in dispersion errors for various propagation angles.

In the case of the lumped mass matrix, there is no the CFL number for which non-dispersive solutions can be obtained for the proposed method and Bathe method (see Fig. 11-12). Therefore, we use $\text{CFL} = 0.5$ for the proposed method and $\text{CFL} = 3$ for the Bathe method. In the case of the central difference method, the considered CFL number is 1, which gives non-

dispersive solutions. A similar trend in the consistent mass matrix is observed – a larger propagation angle results in a lower curve and the proposed method shows a smaller variation in speed errors for various angles (see Fig. 19-21).

For the one-dimensional analysis, we can use the central difference method as all the frequency modes are non-dispersive. As the variation in dispersion errors of this method is larger than that of other methods (see Fig. 21), the central difference method is not suitable for two-dimensional and multidimensional problems. However, as the proposed method has a minimum variation for any mass matrix, it is suitable for wave propagation problems.

2.6 Selection of load magnitude at sub-step

This section discusses how to set the sub-step external forces of the proposed method. The typical modal equation of Eq. (29) is given by the following (see Eq. (49)):

$$\ddot{x} + \omega^2 x = r \quad (49)$$

Using the integration of the time from t_1 to t_2 , we obtain the following (see Eq. (50)):

$$\int_{t_1}^{t_2} \ddot{x} dt = \int_{t_1}^{t_2} (r - \omega^2 x) dt \quad (50)$$

or

$$\int_{t_1}^{t_2} (r - \omega^2 x) dt = {}^{t_2}\dot{x} - {}^{t_1}\dot{x} \quad (51)$$

For the first sub-step, we consider t_1 and t_2 as t and $t + \Delta t/4$, respectively. Applying Eq. (3), Eq. (51) changes as follows (see Eq. (52)):

$$\int_{t_1}^{t_2} (r - \omega^2 x) dt = \frac{\Delta t}{4} {}^t \ddot{x} \quad (52)$$

where the integration is the numerical approximation integration. In this study, the trapezoidal rule is applied to solve the integration. As Eq. (49) is still satisfied for the general loading and trajectory [8], the following equation is obtained in the first sub-step (see Eq. (53)).

$$\int_t^{t+\Delta t/4} r dt = \frac{\Delta t}{4} {}^t r \quad (53)$$

Similarly, for the cases of the second, third, and fourth sub-step, we obtain the following (see Eq. (54)-(56)):

$$\int_{t+\Delta t/4}^{t+\Delta t/2} r dt = \frac{\Delta t}{4} (-{}^t r + 2^{t+\Delta t/4} \hat{r}) \quad (54)$$

$$\int_{t+\Delta t/2}^{t+3\Delta t/4} r dt = \frac{\Delta t}{4} ({}^t r - 2^{t+\Delta t/4} \hat{r} + 2^{t+\Delta t/2} \hat{r}) \quad (55)$$

$$\int_{t+3\Delta t/4}^{t+\Delta t} r dt = \frac{\Delta t}{4} [-(1 + 128\alpha){}^t r + 384\alpha^{t+\Delta t/4} \hat{r} + (2 - 384\alpha)^{t+\Delta t/2} \hat{r} + 128\alpha^{t+3\Delta t/4} \hat{r}] \quad (56)$$

where ${}^{t+\Delta t/4} \hat{r}$, ${}^{t+\Delta t/2} \hat{r}$, and ${}^{t+3\Delta t/4} \hat{r}$ are the modal loads at time $t + \Delta t/4$, $t + \Delta t/2$, and $t + 3\Delta t/4$, respectively, corresponding to the load vectors ${}^{t+\Delta t/4} \hat{\mathbf{R}}$, ${}^{t+\Delta t/2} \hat{\mathbf{R}}$, and ${}^{t+3\Delta t/4} \hat{\mathbf{R}}$, respectively.

From Eq. (53)-(56), the proposed method uses the following Eq. (57)-(58).

$$\int_t^{t+\Delta t} r \, dt = \Delta t \left[-32\alpha^t r + 96\alpha^{t+\Delta t/4} \hat{r} - (96\alpha - 1)^{t+\Delta t/2} \hat{r} + 32\alpha^{t+3\Delta t/4} \hat{r} \right] \quad (57)$$

$$\int_t^{t+\Delta t/2} r \, dt = \frac{\Delta t}{4} t+\Delta t/4 \hat{r} \quad (58)$$

When ${}^t r$ and ${}^{t+\Delta t} r$ at time t and $t + \Delta t$, respectively, are given, then the best values of each sub-step are obtained by the following (see Eq. (59)-(61)):

$${}^{t+\Delta t/4} \hat{r} = \frac{1}{2} ({}^t r + {}^{t+\Delta t/2} \hat{r}) = \frac{1}{4} (3{}^t r + {}^{t+\Delta t} r) \quad (59)$$

$${}^{t+\Delta t/2} \hat{r} = \frac{1}{2} ({}^{t+\Delta t/4} \hat{r} + {}^{t+3\Delta t/4} \hat{r}) = \frac{1}{2} ({}^t r + {}^{t+\Delta t} r) \quad (60)$$

$${}^{t+3\Delta t/4} \hat{r} = \frac{1}{2} ({}^{t+\Delta t/2} \hat{r} + {}^{t+\Delta t} r) = \frac{1}{4} ({}^t r + 3{}^{t+\Delta t} r) \quad (61)$$

The best values of the modal loads correspond to the mean value or the value acquired by the trapezoidal rule. If the actual modal loads at each sub-step are unknown, we can use the best value (see Eq. (59)-(61)). Obviously, to improve the accuracy of the solution, the modal loads can be set to the actual value at each time point.

Based on the results in Sections 2.1, 2.2, and 2.6, we can now summarize a procedure for the proposed four sub-step explicit time integration scheme (see Table 1). In this method, we can use both consistent mass and lumped mass matrices. However, the dispersion error of the consistent mass has a smaller value as compared to that of the lumped mass.

3. Wave propagation solutions

In order to test the non-oscillatory properties of the proposed method, some wave propagation problems are considered. First, we solve the one-dimensional elastic bar impact problems and focus on the numerical oscillations that depend on the mass matrix and CFL number. We then solve the two-dimensional scalar wave propagation problem and focus on the influence of the propagation angle on the numerical oscillations.

In this section, the considered CFL numbers for the proposed four sub-step explicit scheme are 0.22 (consistent mass) and 0.5 (lumped mass), those for the Bathe method are 1 (consistent mass) and 3 (lumped mass), and those for the central difference method are 0.1 (consistent mass) and 1 (lumped mass).

3.1 1D elastic bar impact

The one-dimensional impact problem is a good benchmark problem to test new numerical methods for wave propagation solutions (see Fig. 22) [5,6,35]. In the one-dimensional elastic bar, we compare the solutions of the wave propagation by using the proposed four sub-step explicit scheme, Bathe method, and central difference method. In this section, both consistent mass and lumped mass matrices are used to solve the problem of wave propagation. As can be seen in Fig. 22, the governing wave equation of an elastic bar with free-fixed boundary condition (i.e. free condition at $X = 0$ and fixed condition at $X = L$) is expressed by as follows (see Eq. (62)):

$$\frac{\partial^2 u}{\partial X^2} = \frac{1}{c_0^2} \frac{\partial^2 u}{\partial t^2} \quad (62)$$

In this study, we assume that the length of bar L is 10 m, density ρ is $8 \times 10^3 \text{ kg/m}^3$, elastic modulus E is 200 GPa, and section area A is 1 mm^2 . In the one-dimensional bar, the exact wave speed c_0 amounts to $\sqrt{E/\rho} = 5 \times 10^3 \text{ mm/ms}$. For the finite element analysis, the number of elements is 200, i.e. the length of each element is 0.05 m.

3.1.1 Pulse wave of elastic bar

We consider the velocity $\dot{u}(t)$ at $X = L$ of Fig. 22 as follows (see Eq. (63)):

$$\dot{u}(t) = \begin{cases} 1 \text{ m/s} & t > 0 \\ 0 & t = 0 \end{cases} \quad (63)$$

Using the initial conditions from Eq. (63), the wave of elastic bar is a pulse wave. Fig. 23-24 show the exact solution and the three numerical solutions with consistent and lumped mass matrices. In the proposed method, there are no numerical oscillations regardless of mass matrix, whereas the Bathe method and central difference method cause numerical oscillations when consistent and lumped mass matrices are used. The use of the consistent mass matrix can reduce the numerical oscillations in the Bathe method (see Fig. 23-24). When the lumped matrix is used in the Bathe method ($\text{CFL} = 3$), as the numerical wave speed of the participating wave modes is smaller than the exact wave speed (see Fig. 12), the errors occur backward of the wave front. Considering the central difference method, the numerical

oscillations are observed regardless of the mass matrix. Using the consistent mass matrix for the central difference method (CFL = 0.1), the numerical wave speed that is larger than the exact wave speed results in oscillation forward of the wave front.

3.1.2 Reflection of elastic waves at fixed and free end

When the wave front reaches the end of the bar, the wave is reflected from the end. The boundary conditions affect the phase of the reflective wave. As compared to the velocity of the incident wave at the fixed end, the velocity of the reflective wave has a phase difference of 180°, while the phase of the reflective velocity has the same as the incident velocity at the free end [1]. To further explore the reflection of an elastic wave, the initial conditions are as shown in the following equation (see Eq. (64)):

$$\dot{u}(t) = \begin{cases} 1\text{m/s} & 0 < t < 1\text{ms} \\ 0 & \text{elsewhere} \end{cases} \quad (64)$$

With length of the bar $L = 10$ m, the time required by the wave front to reach the fixed end is 2 ms. The reflective wave occurs when the front of the wave reaches the end of the bar and thereafter the incident and reflective waves are superposed. Fig. 25-26 and 27-28 show the velocity distributions at $t = 2.5$ ms and $t = 4.5$ ms, respectively. In the exact solution, all velocities are 0 at $t = 2.5$ ms and the amplitude of velocities is -2 m/s at $t = 4.5$ ms, as the exact solution is acquired by the principle of superposition. When the proposed four sub-step explicit scheme is performed, the amplitude of numerical oscillations is small as compared to those in other methods, for any mass matrix (see Fig. 25-28).

In Fig. 29, the computational efforts of three methods for a one-dimensional reflection problem are compared. For the consistent mass, the central difference method costs more time than the proposed method as the proposed method has a CFL number fifth that of the central difference method. However, for the lumped mass cases, the proposed method with four sub-steps requires more time than the other methods. Therefore, in the one-dimensional problems, the proposed method requires a substantial amount of time to arrive at a solution despite the solution being non-oscillatory.

3.2 2D scalar wave propagation

The pre-stressed membrane problem is considered for testing the proposed method (see Fig. 30). The governing equation for this problem is expressed as follows (see Eq. (65)) [7,8]:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + F(0,0,t) = \frac{1}{c_0^2} \frac{\partial^2 u}{\partial t^2} \quad (65)$$

where u is the transverse displacement and we assume that the exact wave speed c_0 is 1.0. The impulse load is given by Eq. (66).

$$F(0,0,t) = \begin{cases} 4 - t & 0 < t < 4 \\ 0 & \text{elsewhere} \end{cases} \quad (66)$$

The exact solution is obtained from the following equation (see Eq. (67)) [40]:

$$u(x,y,t) = \int_0^t f(\tau) G(x,y,t-\tau) d\tau \quad (67)$$

where $G(x, y, t)$ is the Green's function, which is given as follows (see Eq. (68)):

$$G(x, y, t) = \frac{H(c_0 t - \sqrt{x^2 + y^2})}{2\pi c_0 \sqrt{c_0^2 t^2 - x^2 - y^2}} \quad (68)$$

where H is the unit step function. As the membrane is symmetric, we only consider the domain $[0, 11]$ by $[0, 11]$. For the CFL number, the characteristic length is the length of the side of the elements.

Fig. 31-32 show the numerical results for the displacement and velocity using an 88 by 88 element mesh and various propagation angles at time $t = 6$. In Fig. 31, the consistent mass matrix is considered, while Fig. 32 shows the results of the lumped mass matrix. The angles used are as 0 and $\pi/4$. The proposed method and Bathe method have reasonably accurate solutions of displacement u . However, the proposed method, Bathe method, and central difference method have significant solution errors in the peak value of the velocity $\dot{u}$. For the various propagation angles, the results of the proposed method are non-oscillatory solutions of displacement and velocity, while both the Bathe method and central difference method cause some numerical oscillations.

In the case of the lumped mass matrix, only the proposed method has accurate and non-oscillatory solutions of displacement. In the numerical results of the Bathe method and central difference method, as the relative wave speed errors are negative (see Fig. 20-21), numerical oscillation errors in the predicted displacement occur behind the wave front.

Fig. 33-38 show snapshots of the displacements at time $t = 6$ for various mesh and mass matrices using the proposed method, Bathe method, and central difference method. We observe that the use of the Bathe method (lumped mass matrix) and central difference method (any mass matrix) causes the

occurrence of numerical oscillations even when using a 176 by 176 mesh. However, the proposed method (any mass matrix) and Bathe method (consistent mass matrix) provide a reasonably accurate solution with an 88 by 88 element mesh. Moreover, in the case of a low spatial discretization density (e.g., 44 by 44 element mesh), only the proposed method provides non-oscillatory numerical results.

Fig. 39 shows a rough quantitative comparison of the computational efforts required when using three numerical simulation methods. In the multidimensional problems, the bandwidths of the mass and stiffness matrices are greater than those of the one-dimensional problems. The proposed method requires less effort for the calculation matrix (see Table. 1), while the Bathe method and central difference method require more matrix calculation than the proposed method. Therefore, although the proposed method comprises four sub-steps, this method requires slightly less time than other methods unlike in the one-dimensional reflection problem. Therefore, the proposed method is desirable for multi-dimensional problems owing to its non-oscillatory solutions and acceptable time cost.

4. Concluding remarks

In the present study, a new four sub-step explicit time integration scheme is proposed for the analysis of stress wave propagation problems. In the proposed method, considering the fourth-order central difference method and a high-order derivative term, there are four sub-steps. The high-order derivative term affects the stability limit. Using the appropriate value of parameter α , a better stability than the central difference method is obtained. The parameter $\alpha = 0.045$ is recommended for maximum stability limit, minimum dissipation, and accuracy. We also propose the best values of the modal loads at each sub-step.

Furthermore, a dispersion analysis for the mass matrices is performed. When the consistent matrix with $CFL = 0.22$ is used, the effective wave modes are non-dispersive. In the case of the lumped mass matrix, there is no CFL number when the participating wave modes are non-dispersive. Therefore, as the maximum absolute value of the relative wave speed error is ca. 0.02, we recommend using $CFL = 0.5$ for the lumped mass.

We then present the performances of the proposed method, Bathe method, and central difference method in the solution of a one-dimensional elastic bar and a two-dimensional scalar wave propagation problem and compare the numerical solutions with the exact solution. Of note, considering the propagation and reflection of waves, there are no numerical oscillations for the proposed method, while the Bathe method and central difference method cause numerical oscillations. The Bathe method with consistent mass has a low numerical oscillation, whereas that with lumped mass yields large errors and numerical oscillations. Since the computational effort in the consistent mass matrix is larger than that required in the lumped mass matrix, and the proposed method with lumped mass has no oscillations, it can be used to decrease computational cost. Moreover, considering that the oscillation amplitude of the proposed method is smaller than those of other methods, as compared to the results of the wave reflection, the proposed method can be used to reduce the numerical oscillations.

The proposed method has both advantages and disadvantages. At the same value of the modal natural frequency, the critical time step of proposed method is larger than that of central difference method, which is a widely used explicit scheme (the proposed method thus has a good stability). As shown by the results of the numerical solutions, the proposed method with consistent and lumped mass matrices yields non-oscillatory results for wave propagation problems. On the other hand, the proposed method has a small CFL number that results into the smallest dispersion error. In addition, the number of sub-steps in

this method is four, which is larger than in other general time integration schemes. This suggests a large time increment and implies that the proposed method requires a large amount of computational time/cost. However, as the proposed method has a lower cost of calculating the matrix, the proposed method is desirable for two-dimensional and three-dimensional problems.

For the stress wave problem, the constitutive relation is also important. In this study, we consider the constitutive relation as Hook's law, owing to elastic wave propagation. However, the numerical solution can vary depending on the constitutive model when the problems of plastic wave propagation are solved. In the future, theoretical and numerical studies on the combination of the time integral scheme and constitutive models, such as the Gurson, Frank-Brockman, TI (Tomita/Iwamoto), BC (Bodner/Chan), and BP (Bodner/Partom) models, would be useful for the analysis of the propagation of stress waves.

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Appendix A. The integration and load operators

The integration operator $\mathbf{A}$ and load operators $\mathbf{L}_a$, $\mathbf{L}_b$, $\mathbf{L}_c$, and $\mathbf{L}_d$ are calculated using the following equations (Eq. (69)-(73)):

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad (69)$$

$$\mathbf{L}_a = \begin{bmatrix} L_{a1} \\ L_{a2} \\ L_{a3} \end{bmatrix} \quad (70)$$

$$\mathbf{L}_b = \begin{bmatrix} \Omega_0[8\alpha\xi\Omega_0^2 + (64\alpha\xi^2 - 16\alpha - 1)\Omega_0 + 4(96\alpha - 1)] \\ -\Delta t[4\alpha\Omega_0^2 + 32\alpha\xi\Omega_0 + 96\alpha - 1] \\ (\Delta t)^2[16\alpha + 1] \end{bmatrix} \quad (71)$$

$$\mathbf{L}_c = \begin{bmatrix} -64\alpha\xi\Omega_0 \\ 32\alpha\Delta t \\ 0 \end{bmatrix} \quad (72)$$

$$\mathbf{L}_d = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (73)$$

where, as shown in Eq. (74)-(85),

$$a_{11} = \frac{\Omega_0}{256} \begin{bmatrix} 8\alpha\xi\Omega_0^6 + (256\alpha\xi^2 - 16\alpha - 1)\Omega_0^5 + 2\xi(1280\alpha\xi^2 - 288\alpha - 13)\Omega_0^4 \\ +4(2048\alpha\xi^4 - 4(96\alpha + 11)\xi^2 + 32\alpha + 9)\Omega_0^3 \\ +128\xi(2(96\alpha - 1)\xi^2 - 80\alpha + 1)\Omega_0^2 \\ +8192\alpha(4\xi^2 - 1)\Omega_0 + 16384\alpha\xi \end{bmatrix} \quad (74)$$

$$a_{12} = \frac{\Omega_0}{32\Delta t} \begin{bmatrix} 8\alpha\xi\Omega_0^6 + (192\alpha\xi^2 - 16\alpha - 1)\Omega_0^5 + 2\xi(768\alpha\xi^2 - 160\alpha - 9)\Omega_0^4 \\ +4(1024\alpha\xi^4 - 8(16\alpha + 3)\xi^2 - 32\alpha + 5)\Omega_0^3 \\ +128\xi((64\alpha - 1)\xi^2 - 16\alpha + 1)\Omega_0^2 \\ +32(4(32\alpha + 1)\xi^2 - 1)\Omega_0 - 64\xi \end{bmatrix} \quad (75)$$

$$a_{13} = \frac{\Omega_0}{8(\Delta t)^2} \begin{bmatrix} 8\alpha\xi\Omega_0^5 + (128\alpha\xi^2 - 16\alpha - 1)\Omega_0^4 + 2\xi(256\alpha\xi^2 - 64\alpha - 5)\Omega_0^3 \\ +4(4(64\alpha - 1)\xi^2 - 64\alpha + 3)\Omega_0^2 + 16\xi(32\alpha + 1)\Omega_0 - 8 \end{bmatrix} \quad (76)$$

$$a_{21} = -\frac{\Delta t}{256} \begin{bmatrix} 4\alpha\Omega_0^6 + 128\alpha\xi\Omega_0^5 + (1280\alpha\xi^2 - 96\alpha - 1)\Omega_0^4 \\ +8\xi(512\alpha\xi^2 + 32\alpha - 3)\Omega_0^3 + 32(4(96\alpha - 1)\xi^2 - 64\alpha + 1)\Omega_0^2 \\ +16384\alpha\xi\Omega_0 + 8192\alpha \end{bmatrix} \quad (77)$$

$$a_{22} = -\frac{1}{32} \left[\begin{aligned} &4\alpha\Omega_0^6 + 96\alpha\xi\Omega_0^5 + (768\alpha\xi^2 - 32\alpha - 1)\Omega_0^4 \\ &+ 16\xi(128\alpha\xi^2 + 16\alpha - 1)\Omega_0^3 + 16(4(64\alpha - 1)\xi^2 + 1)\Omega_0^2 \\ &+ 64\xi(32\alpha + 1)\Omega_0 - 32 \end{aligned} \right] \quad (78)$$

$$a_{23} = -\frac{\Omega_0^2}{8\Delta t} [4\alpha\Omega_0^4 + 64\alpha\xi\Omega_0^3 + (256\alpha\xi^2 - 1)\Omega_0^2 + 8\xi(64\alpha - 1)\Omega_0 + 8(32\alpha + 1)] \quad (79)$$

$$a_{31} = \frac{(\Delta t)^2}{256} \left[\begin{aligned} &(16\alpha + 1)\Omega_0^4 + 24\xi(16\alpha + 1)\Omega_0^3 + 4(32(16\alpha + 1)\xi^2 - 32\alpha - 9)\Omega_0^2 \\ &+ 64\xi(96\alpha - 1)\Omega_0 + 8192\alpha \end{aligned} \right] \quad (80)$$

$$a_{32} = \frac{\Delta t}{32} \left[\begin{aligned} &(16\alpha + 1)\Omega_0^4 + 16\xi(16\alpha + 1)\Omega_0^3 \\ &+ 4(16(16\alpha + 1)\xi^2 + 32\alpha - 5)\Omega_0^2 + 32\xi(64\alpha - 3)\Omega_0 + 32 \end{aligned} \right] \quad (81)$$

$$a_{33} = \frac{1}{8} [(16\alpha + 1)\Omega_0^4 + 8\xi(16\alpha + 1)\Omega_0^3 + 4(64\alpha - 3)\Omega_0^2 + 8] \quad (82)$$

$$L_{a1} = -\frac{\Omega_0}{8} \left[\begin{aligned} &8\alpha\xi\Omega_0^4 + (128\alpha\xi^2 - 16\alpha - 1)\Omega_0^3 + 2\xi(256\alpha\xi^2 - 32\alpha - 5)\Omega_0^2 \\ &+ 4(4\xi^2 - 1)(96\alpha - 1)\Omega_0 + 1536\alpha\xi \end{aligned} \right] \quad (83)$$

$$L_{a2} = \frac{\Delta t}{8} [4\alpha\Omega_0^4 + 64\alpha\xi\Omega_0^3 + (256\alpha\xi^2 + 32\alpha - 1)\Omega_0^2 + 8\xi(96\alpha - 1)\Omega_0 + 768\alpha] \quad (84)$$

$$L_{a3} = -\frac{(\Delta t)^2}{8} [(16\alpha + 1)\Omega_0^2 + 8\xi(16\alpha + 1)\Omega_0 + 4(96\alpha - 1)] \quad (85)$$

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List of Tables

Table 1 Step-by-step solution using the four sub-step explicit time integration scheme

Table 1 Step-by-step solution using the four sub-step explicit time integration scheme

A. Initial calculations

1. Form mass matrix $\mathbf{M}$, stiffness matrix $\mathbf{K}$, and damping matrix $\mathbf{C}$
2. Initialize ${}^0\mathbf{U}$, ${}^0\dot{\mathbf{U}}$, and ${}^0\ddot{\mathbf{U}}$
3. Select time step Δt , $\Delta t < \Delta t_{cr}$, and parameter α , and calculate the integration constants:

$$a_0 = \frac{\Delta t}{4}; \quad a_1 = \frac{1}{2}a_0^2; \quad a_2 = 2a_0; \quad a_3 = \frac{1}{2}a_2^2; \quad a_4 = -6a_2; \quad a_5 = \alpha(\Delta t)^4;$$

$$p_0 = -\frac{32}{(\Delta t)^3}; \quad p_1 = -3p_0;$$

B. For each time step

<First sub-step>

1. Calculate displacements, velocities, and effective loads at time $t + \Delta t/4$:

$${}^{t+\Delta t/4}\mathbf{U} = {}^t\mathbf{U} + a_0 {}^t\dot{\mathbf{U}} + a_1 {}^t\ddot{\mathbf{U}}$$

$${}^{t+\Delta t/4}\dot{\mathbf{U}} = {}^t\dot{\mathbf{U}} + a_0 {}^t\ddot{\mathbf{U}}$$

$${}^{t+\Delta t/4}\hat{\mathbf{R}} = \frac{1}{4}(3 {}^t\mathbf{R} + {}^{t+\Delta t}\mathbf{R})$$

2. Solve for accelerations at time $t + \Delta t/4$:

$$\mathbf{M}^{t+\Delta t/4}\ddot{\mathbf{U}} = {}^{t+\Delta t/4}\hat{\mathbf{R}} - \mathbf{C}^{t+\Delta t/4}\dot{\mathbf{U}} - \mathbf{K}^{t+\Delta t/4}\mathbf{U}$$

<Second sub-step>

1. Calculate displacements, velocities, and effective loads at time $t + \Delta t/2$:

$${}^{t+\Delta t/2}\mathbf{U} = {}^t\mathbf{U} + a_2 {}^{t+\Delta t/4}\dot{\mathbf{U}} + a_3 {}^{t+\Delta t/4}\ddot{\mathbf{U}}$$

$${}^{t+\Delta t/2}\dot{\mathbf{U}} = {}^t\dot{\mathbf{U}} + a_2 {}^{t+\Delta t/4}\ddot{\mathbf{U}}$$

$${}^{t+\Delta t/2}\hat{\mathbf{R}} = \frac{1}{2}({}^t\mathbf{R} + {}^{t+\Delta t}\mathbf{R})$$

2. Solve for accelerations at time $t + \Delta t/2$:

$$\mathbf{M}^{t+\Delta t/2}\ddot{\mathbf{U}} = {}^{t+\Delta t/2}\hat{\mathbf{R}} - \mathbf{C}^{t+\Delta t/2}\dot{\mathbf{U}} - \mathbf{K}^{t+\Delta t/2}\mathbf{U}$$

<Third sub-step>

1. Calculate displacements, velocities, and effective loads at time $t + 3\Delta t/4$:

$${}^{t+3\Delta t/4}\mathbf{U} = {}^{t+\Delta t/4}\mathbf{U} + a_2 {}^{t+\Delta t/2}\dot{\mathbf{U}} + a_3 {}^{t+\Delta t/2}\ddot{\mathbf{U}}$$

$${}^{t+3\Delta t/4}\dot{\mathbf{U}} = {}^{t+\Delta t/4}\dot{\mathbf{U}} + a_2 {}^{t+\Delta t/2}\ddot{\mathbf{U}}$$

$${}^{t+3\Delta t/4}\hat{\mathbf{R}} = \frac{1}{4}({}^t\mathbf{R} + 3 {}^{t+\Delta t}\mathbf{R})$$

2. Solve for accelerations at time $t + 3\Delta t/4$:

$$\mathbf{M}^{t+3\Delta t/4}\ddot{\mathbf{U}} = {}^{t+3\Delta t/4}\hat{\mathbf{R}} - \mathbf{C}^{t+3\Delta t/4}\dot{\mathbf{U}} - \mathbf{K}^{t+3\Delta t/4}\mathbf{U}$$

<Fourth sub-step>

1. Calculate fourth-order and fifth-order derivative terms at time $t + \Delta t/2$:
-

$${}^{t+\Delta t/2}\mathbf{U}^{(4)} = p_0({}^t\dot{\mathbf{U}} - {}^{t+3\Delta t/4}\dot{\mathbf{U}}) + p_1({}^{t+\Delta t/4}\dot{\mathbf{U}} - {}^{t+\Delta t/2}\dot{\mathbf{U}})$$

$${}^{t+\Delta t/2}\mathbf{U}^{(5)} = p_0({}^t\ddot{\mathbf{U}} - {}^{t+3\Delta t/4}\ddot{\mathbf{U}}) + p_1({}^{t+\Delta t/4}\ddot{\mathbf{U}} - {}^{t+\Delta t/2}\ddot{\mathbf{U}})$$

2. Solve for displacements and velocities at time $t + \Delta t$:

$${}^{t+\Delta t}\mathbf{U} = {}^t\mathbf{U} - 8{}^{t+\Delta t/4}\mathbf{U} + 8{}^{t+3\Delta t/4}\mathbf{U} + a_4{}^{t+\Delta t/2}\dot{\mathbf{U}} + a_5{}^{t+\Delta t/2}\mathbf{U}^{(4)}$$

$${}^{t+\Delta t}\dot{\mathbf{U}} = {}^t\dot{\mathbf{U}} - 8{}^{t+\Delta t/4}\dot{\mathbf{U}} + 8{}^{t+3\Delta t/4}\dot{\mathbf{U}} + a_4{}^{t+\Delta t/2}\ddot{\mathbf{U}} + a_5{}^{t+\Delta t/2}\mathbf{U}^{(5)}$$

3. Calculate accelerations at time $t + \Delta t$:

$$\mathbf{M}{}^{t+\Delta t}\ddot{\mathbf{U}} = {}^{t+\Delta t}\widehat{\mathbf{R}} - \mathbf{C}{}^{t+\Delta t}\dot{\mathbf{U}} - \mathbf{K}{}^{t+\Delta t}\mathbf{U}$$

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Fig. 38 Snapshots of displacement at $t = 6$ for the central difference method, when the lumped mass matrix considered.

Fig. 39 Required computational efforts of various methods considering the 2D wave problem for 88×88 element mesh in section 3.2.

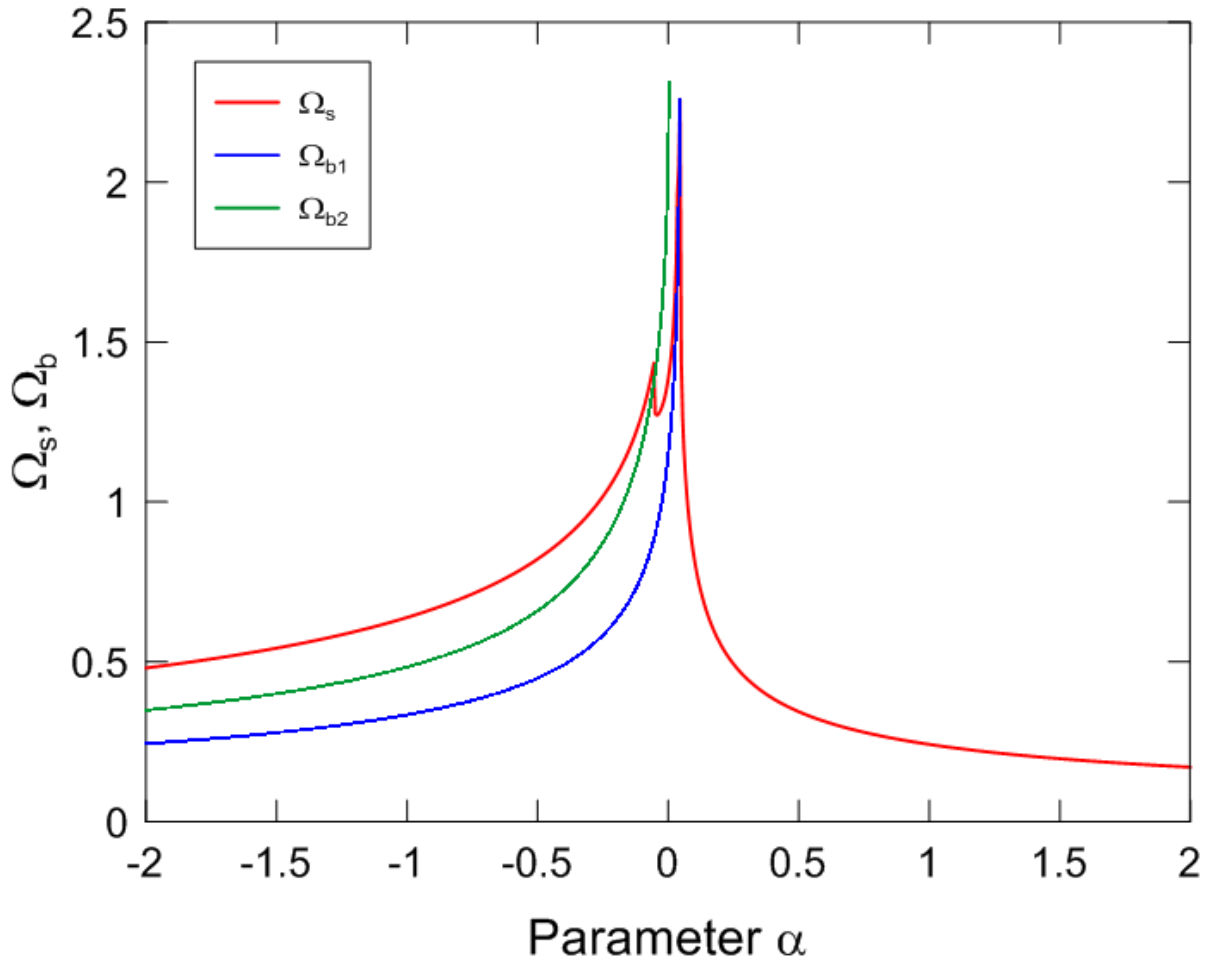


Fig. 1. Stability limit and bifurcation point of the proposed explicit time integration scheme for various values of α .

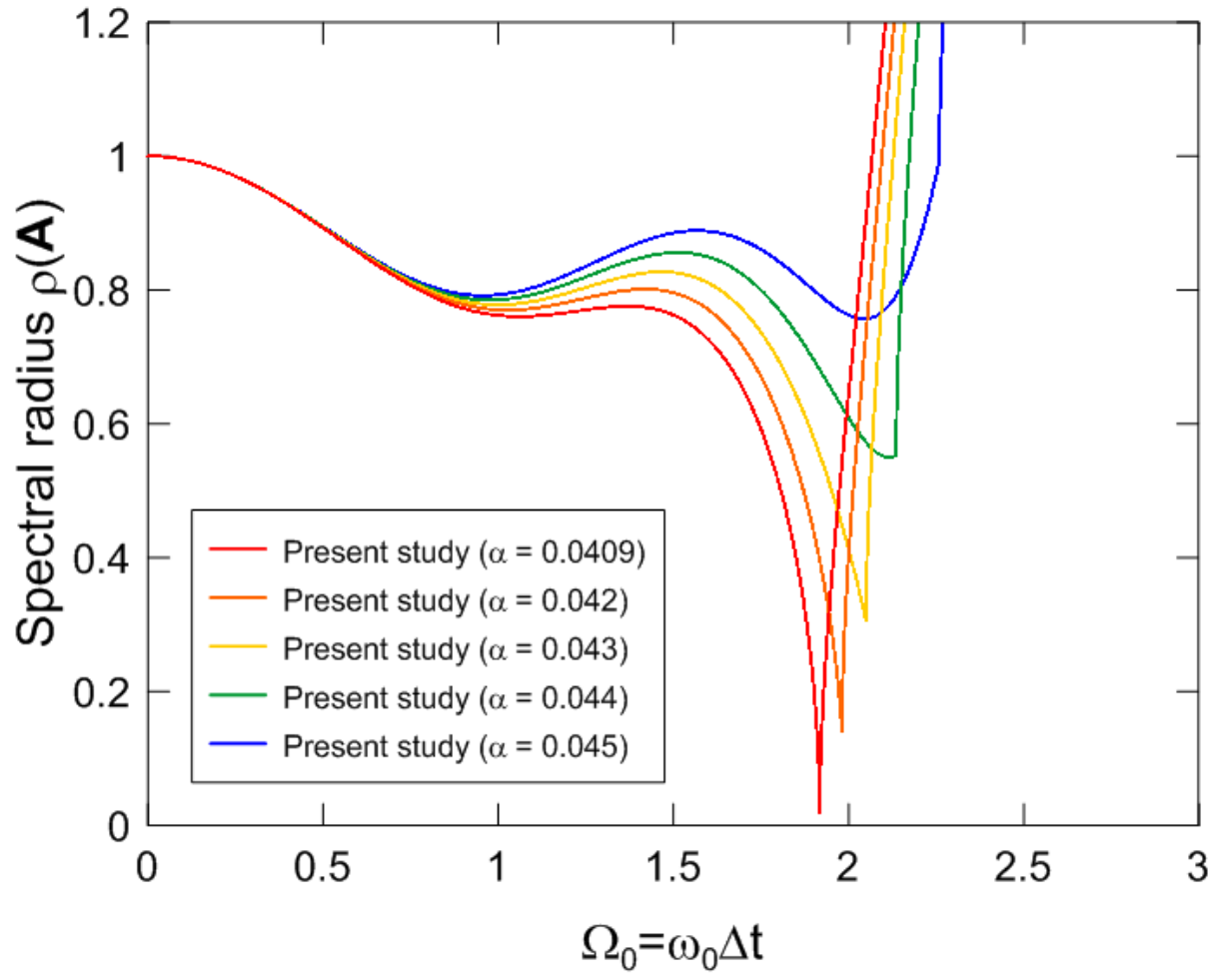


Fig. 2. Spectral radius of the proposed method for various values of α considering no damping.

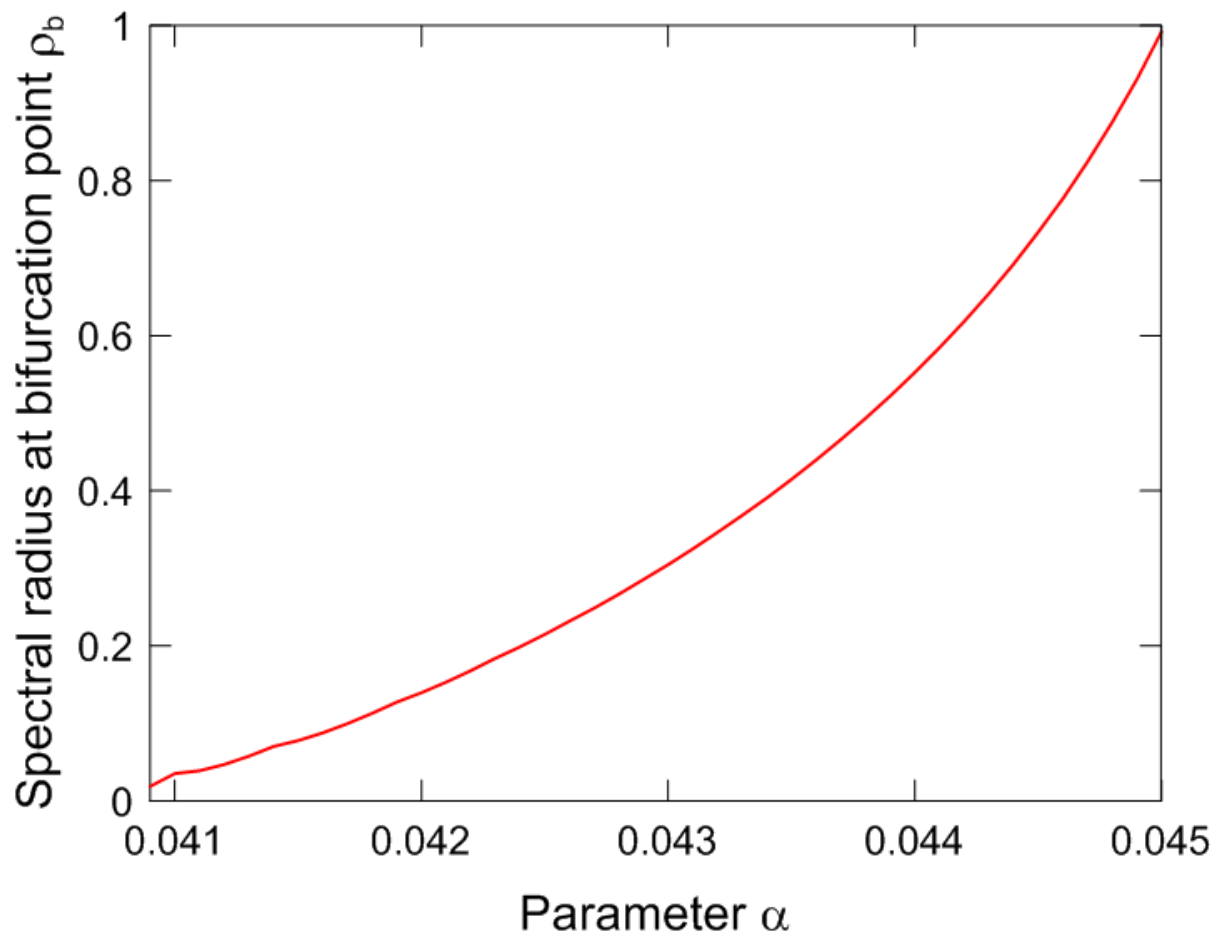


Fig. 3. Spectral radius at the bifurcation point for various values of α considering no damping.

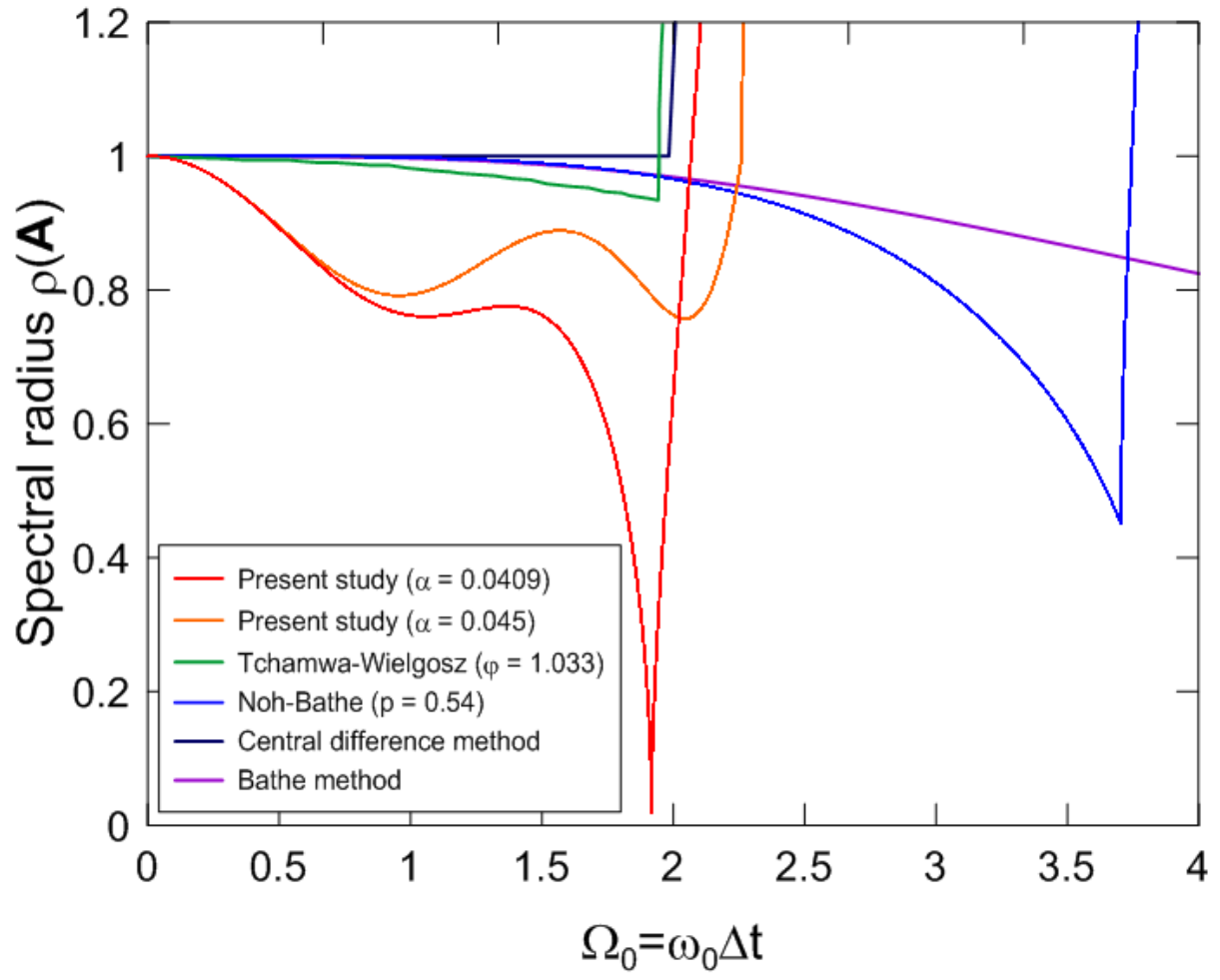
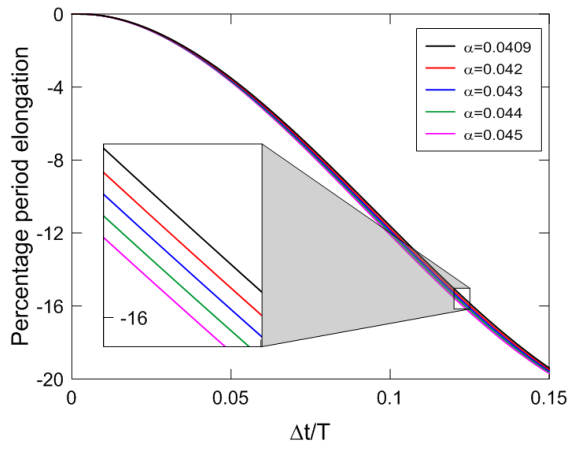
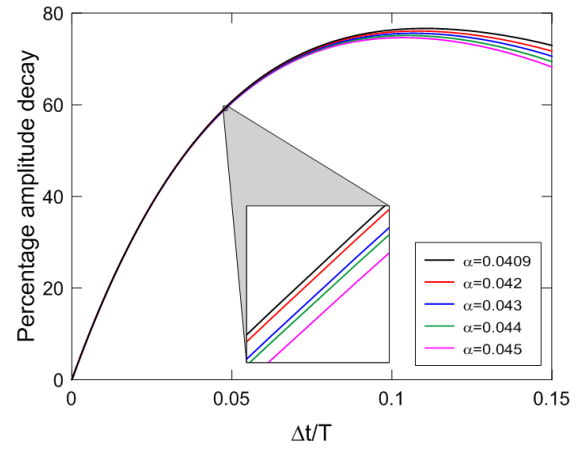


Fig. 4. Spectral radius of the proposed method and other methods considering no damping.

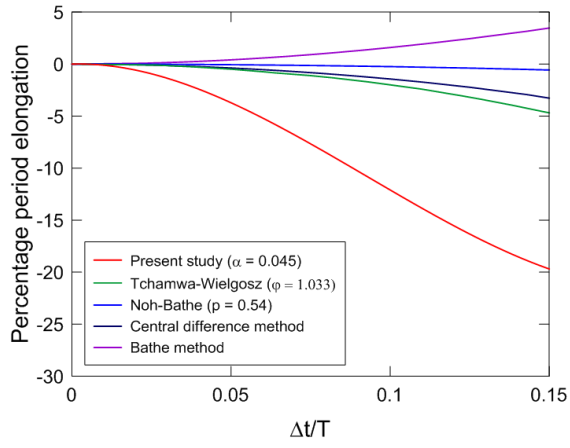


(a) Percentage period elongation

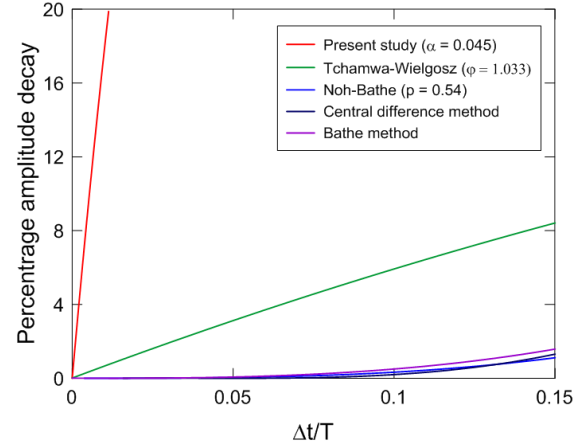


(b) Percentage amplitude decay

Fig. 5. Accuracy analysis of the proposed method for various values of α .



(a) Percentage period elongations



(b) Percentage amplitude decays

Fig. 6. Accuracy analysis of the proposed method ($\alpha = 0.045$) and other methods.

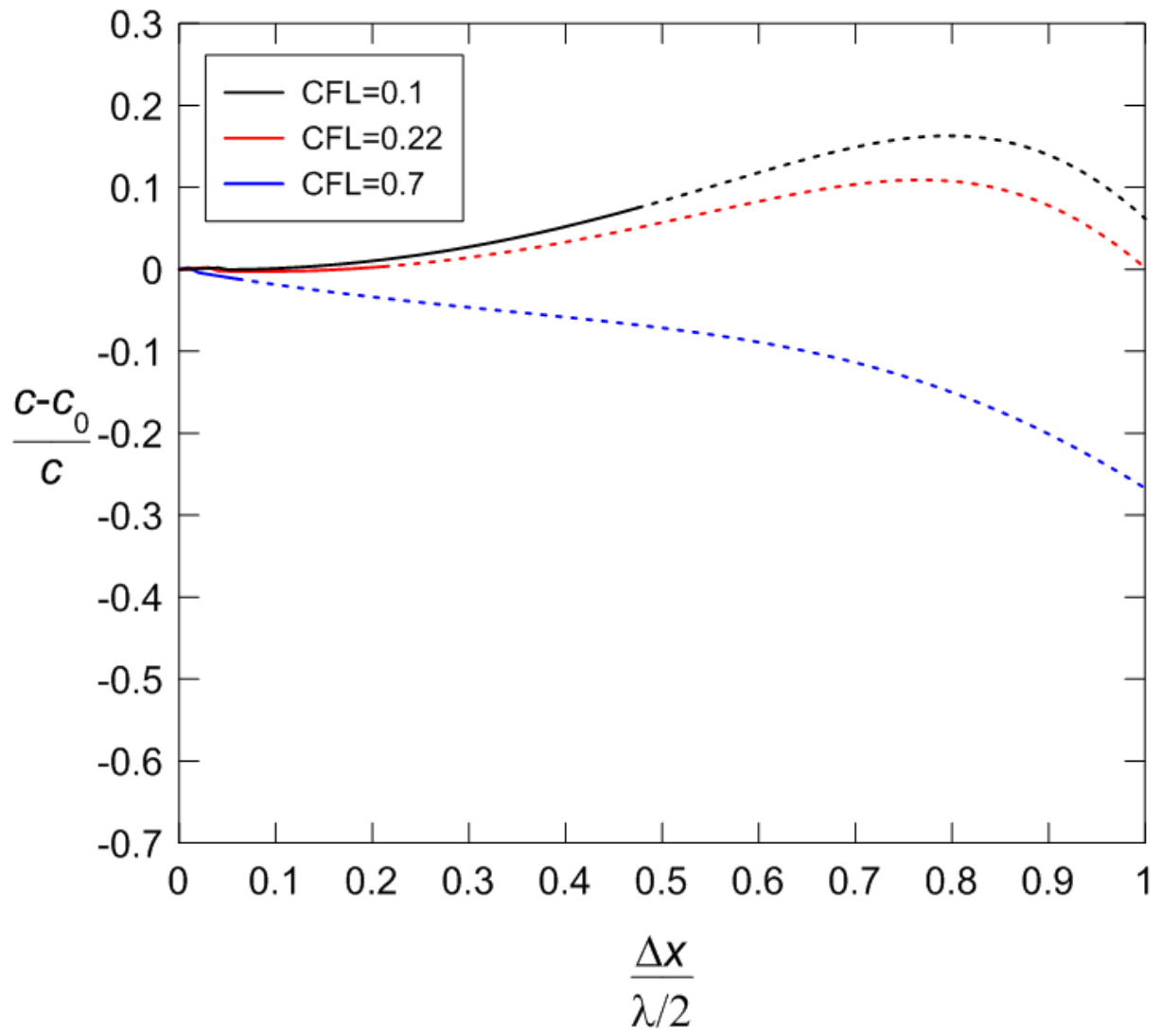


Fig. 7. Relative wave speed errors of the proposed method ($\alpha = 0.045$) for various CFL numbers considering the consistent mass matrix (dotted lines: discarded wave modes).

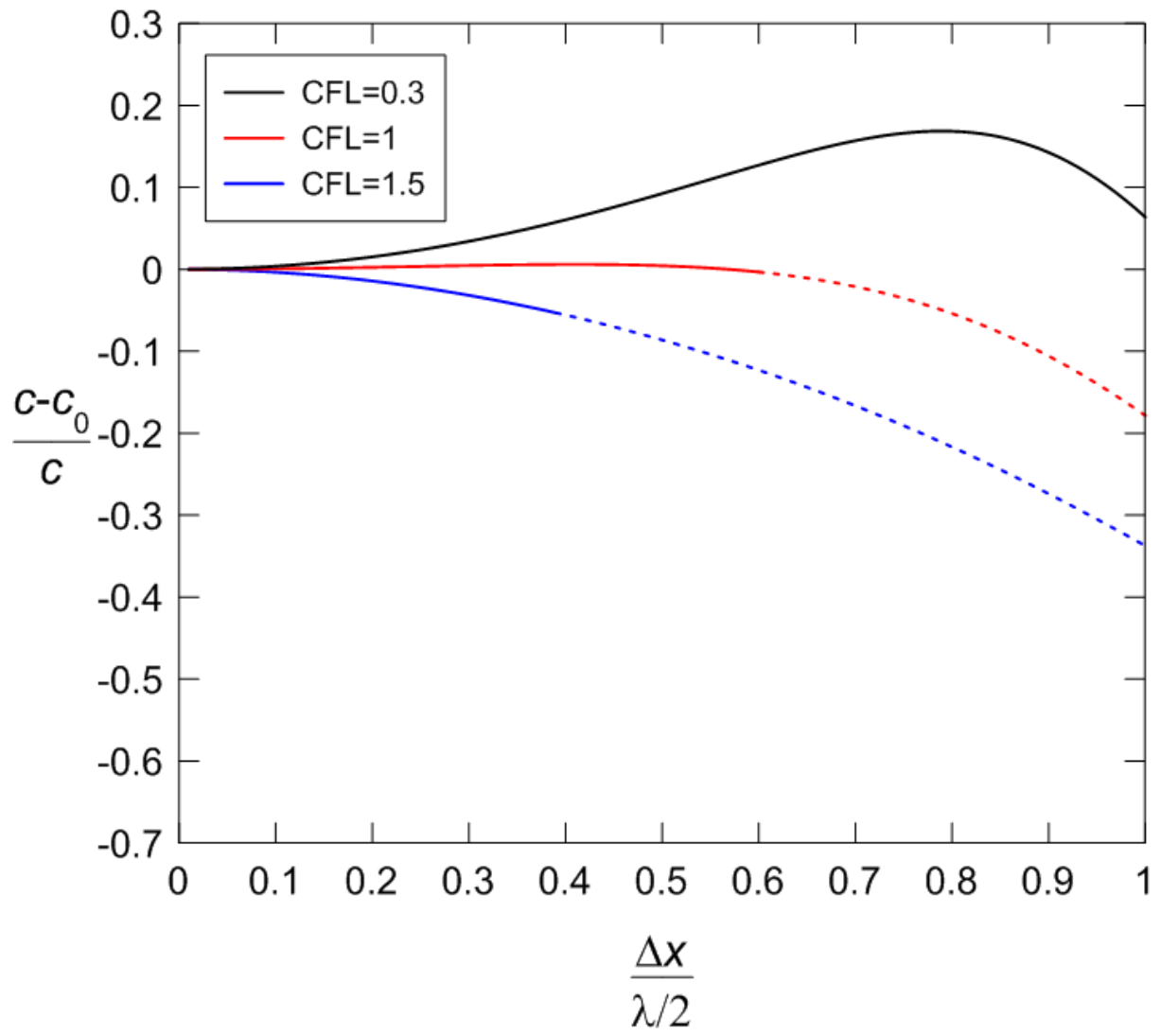


Fig. 8. Relative wave speed errors of the Bathe method for various CFL numbers considering the consistent mass matrix (dotted lines: discarded wave modes).

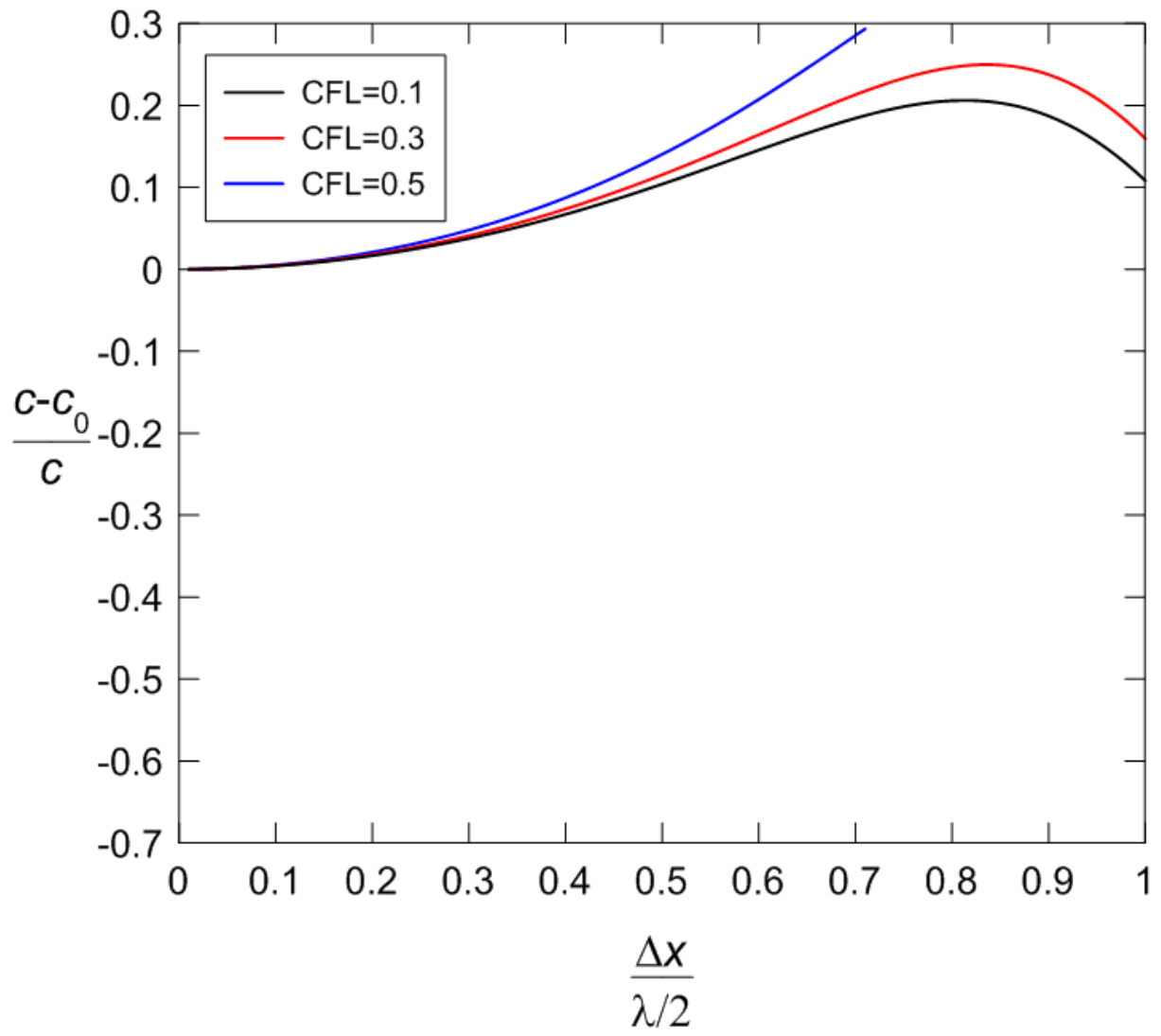


Fig. 9. Relative wave speed errors of the central difference method for various CFL numbers considering the consistent mass matrix.

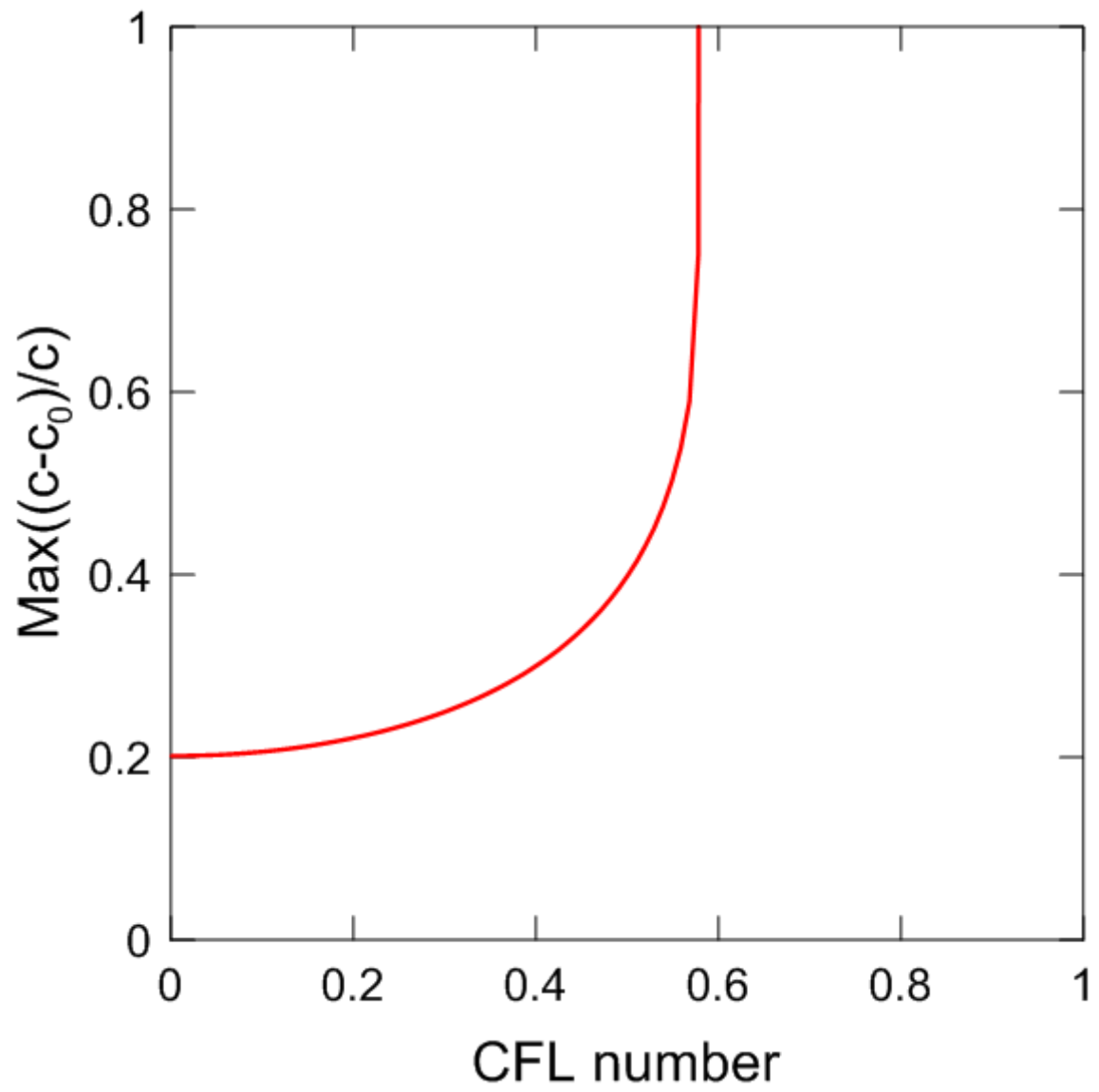


Fig. 10. Convergence rate of relative wave speed errors for the central difference method considering the consistent mass matrix.

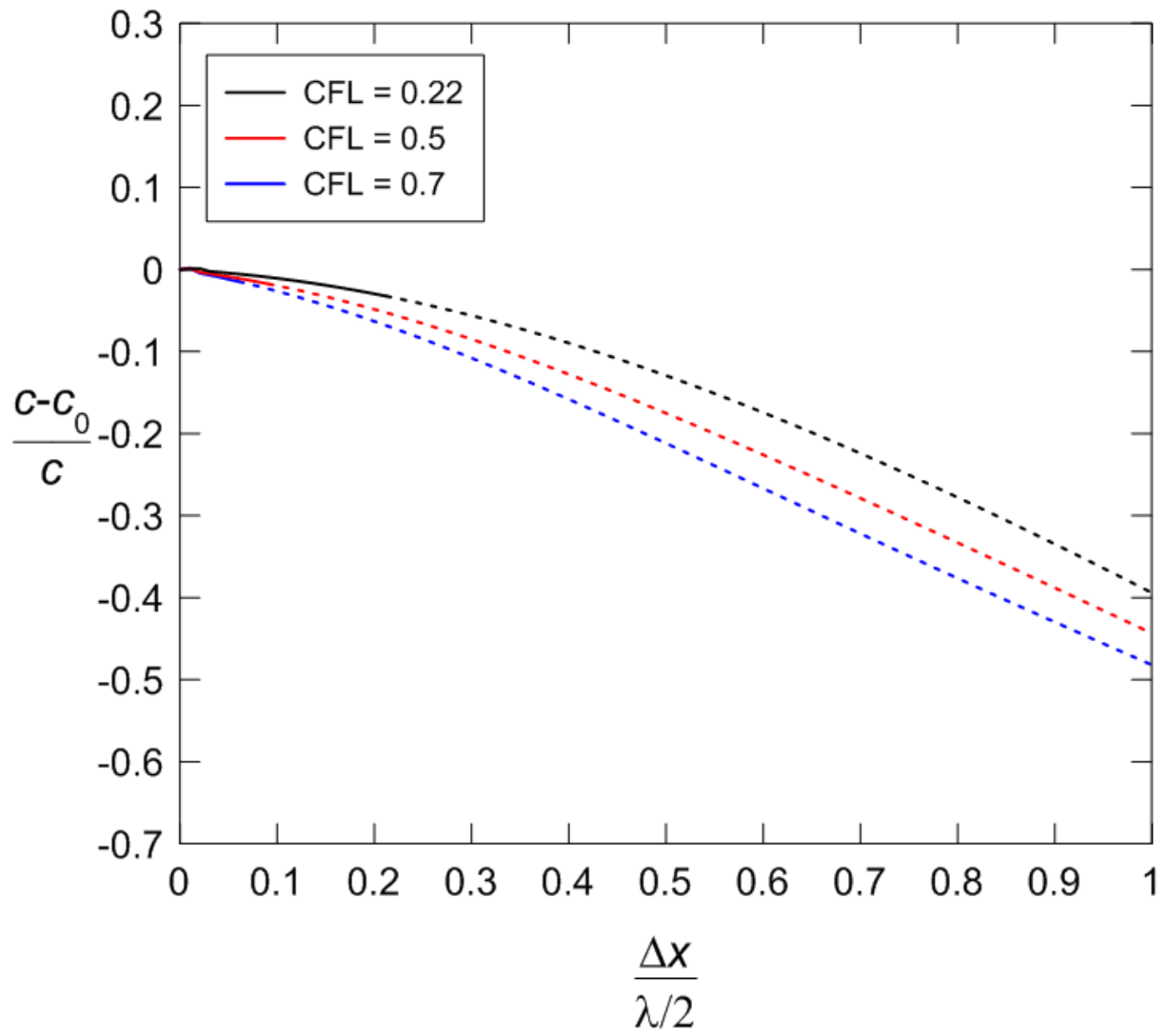


Fig. 11. Relative wave speed errors of the proposed method ($\alpha = 0.045$) for various CFL numbers considering the lumped mass matrix (dotted lines: discarded wave modes).

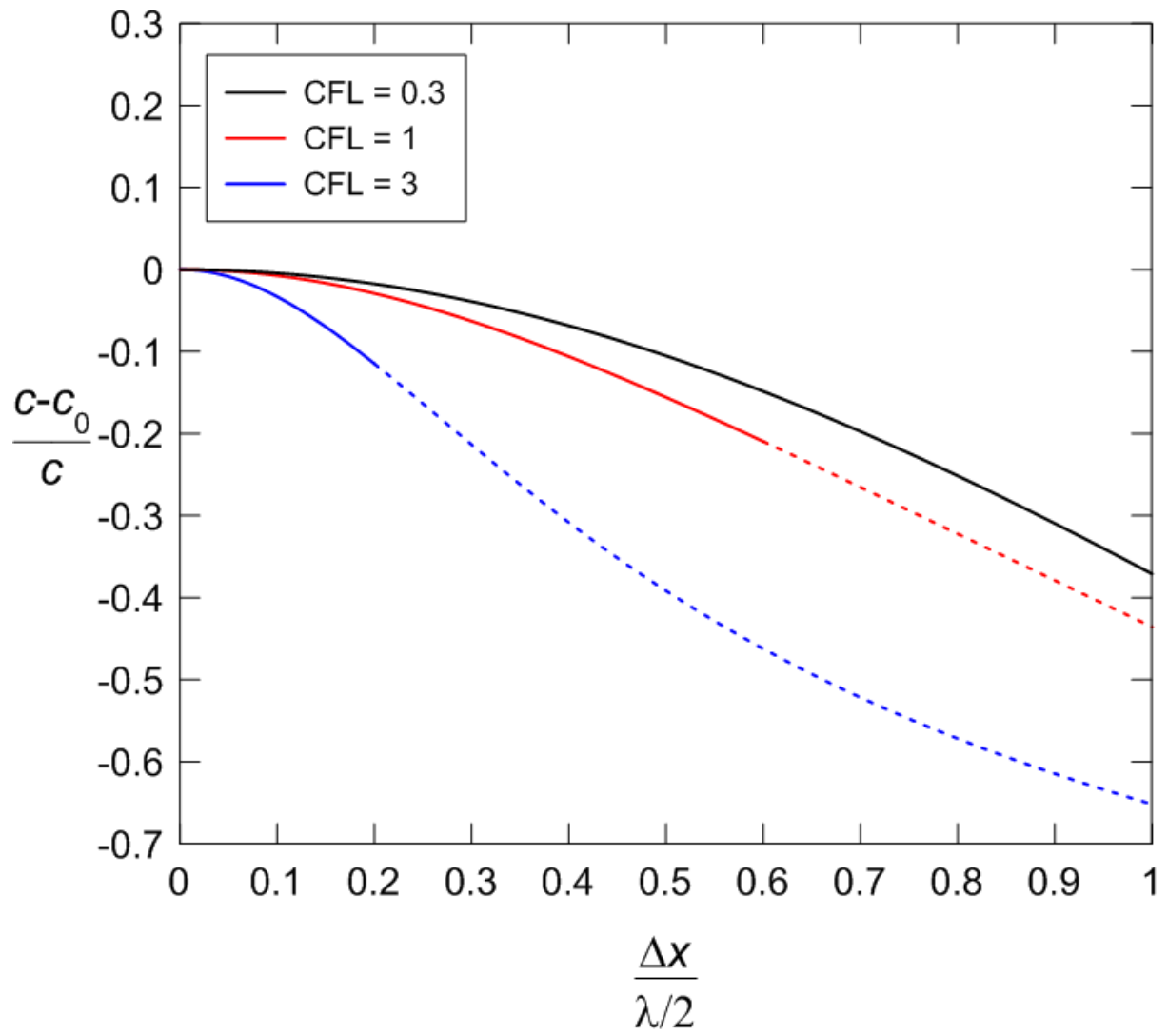


Fig. 12. Relative wave speed errors of the Bathe method for various CFL numbers considering the lumped mass matrix (dotted lines: discarded wave modes).

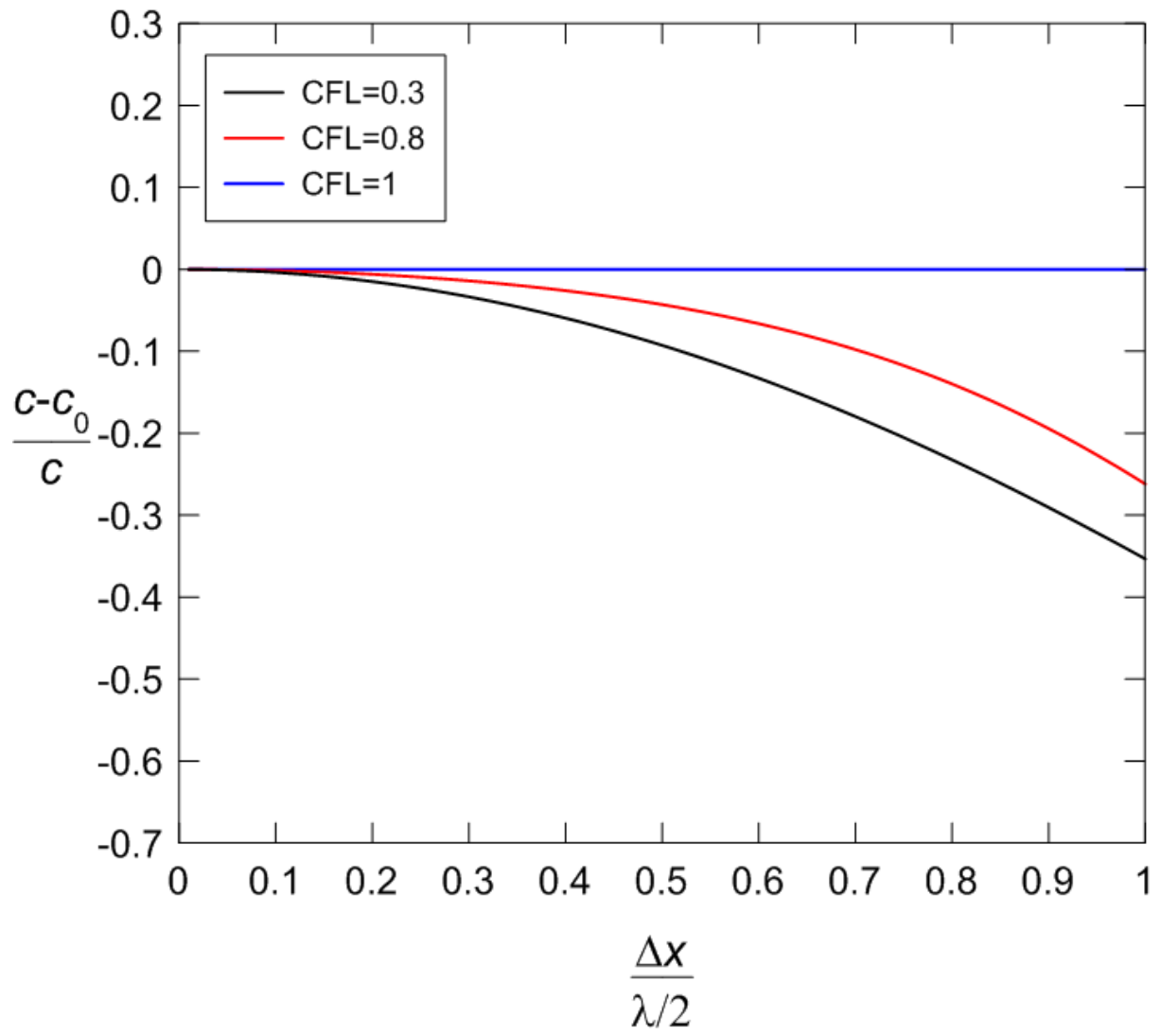


Fig. 13. Relative wave speed errors of the central difference method for various CFL numbers considering the lumped mass matrix.

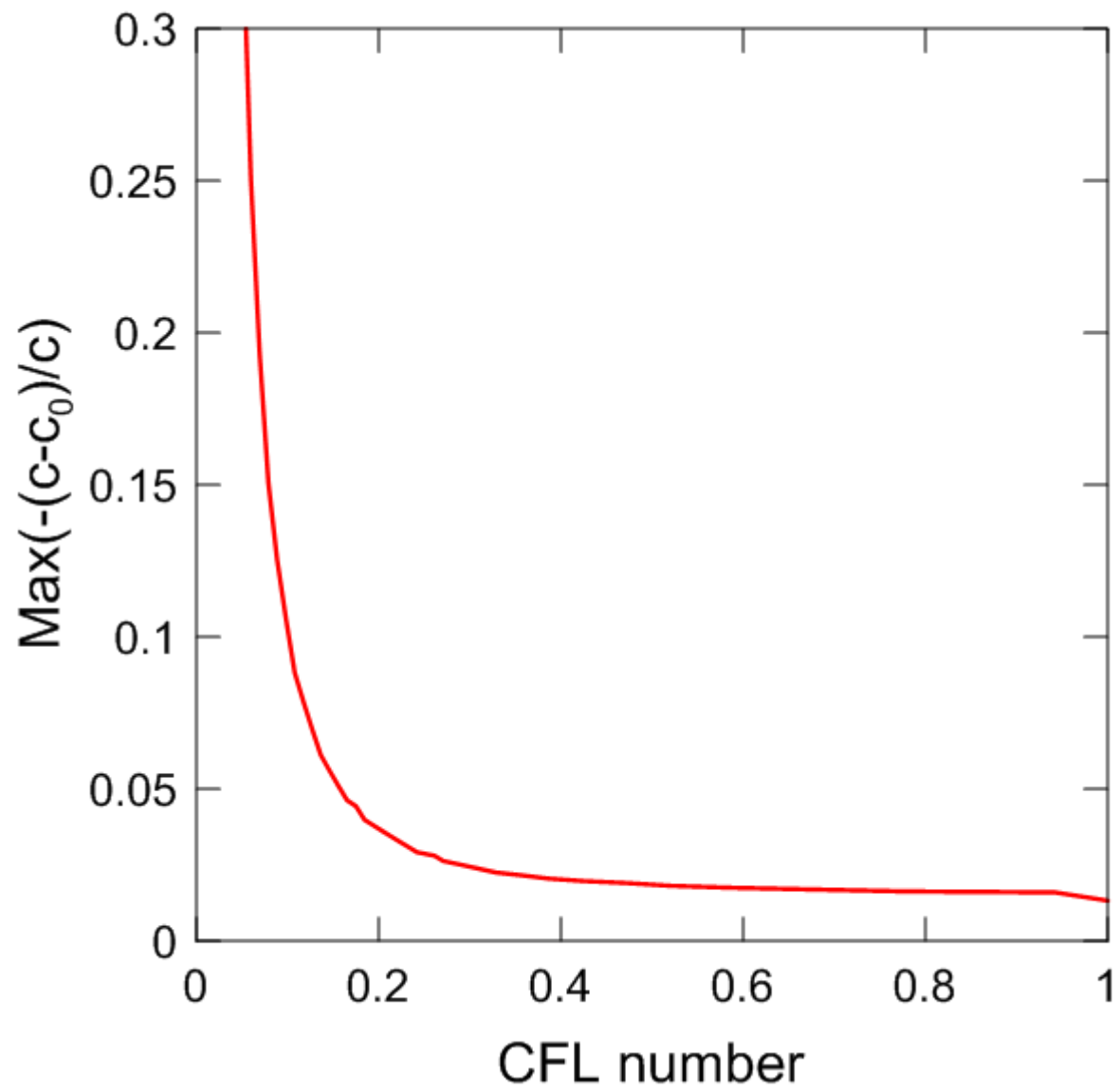


Fig. 14. Convergence rate of relative wave speed errors for the proposed method considering the lumped mass matrix.

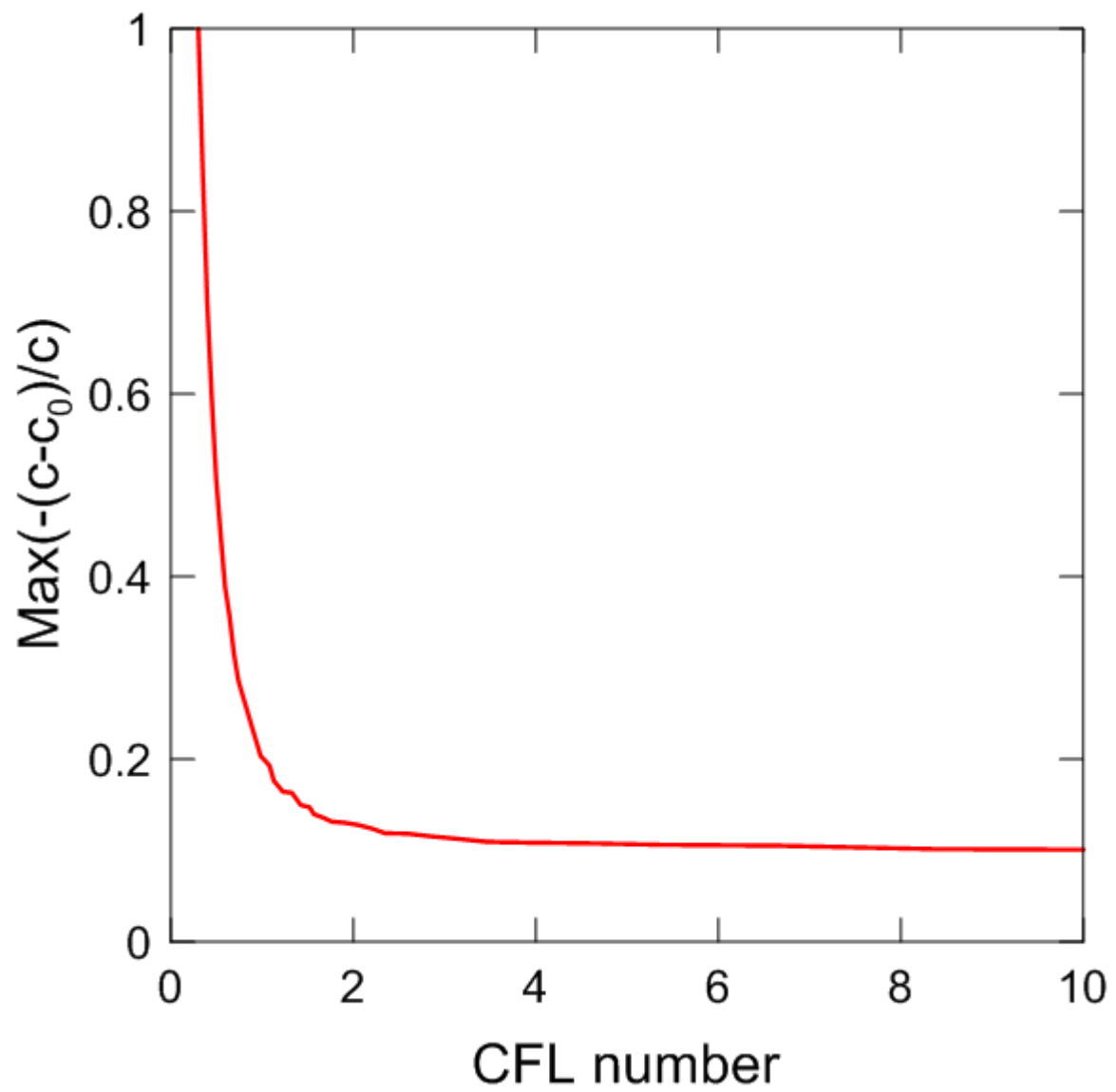


Fig. 15. Convergence rate of relative wave speed errors for the Bathe method considering the lumped mass matrix.

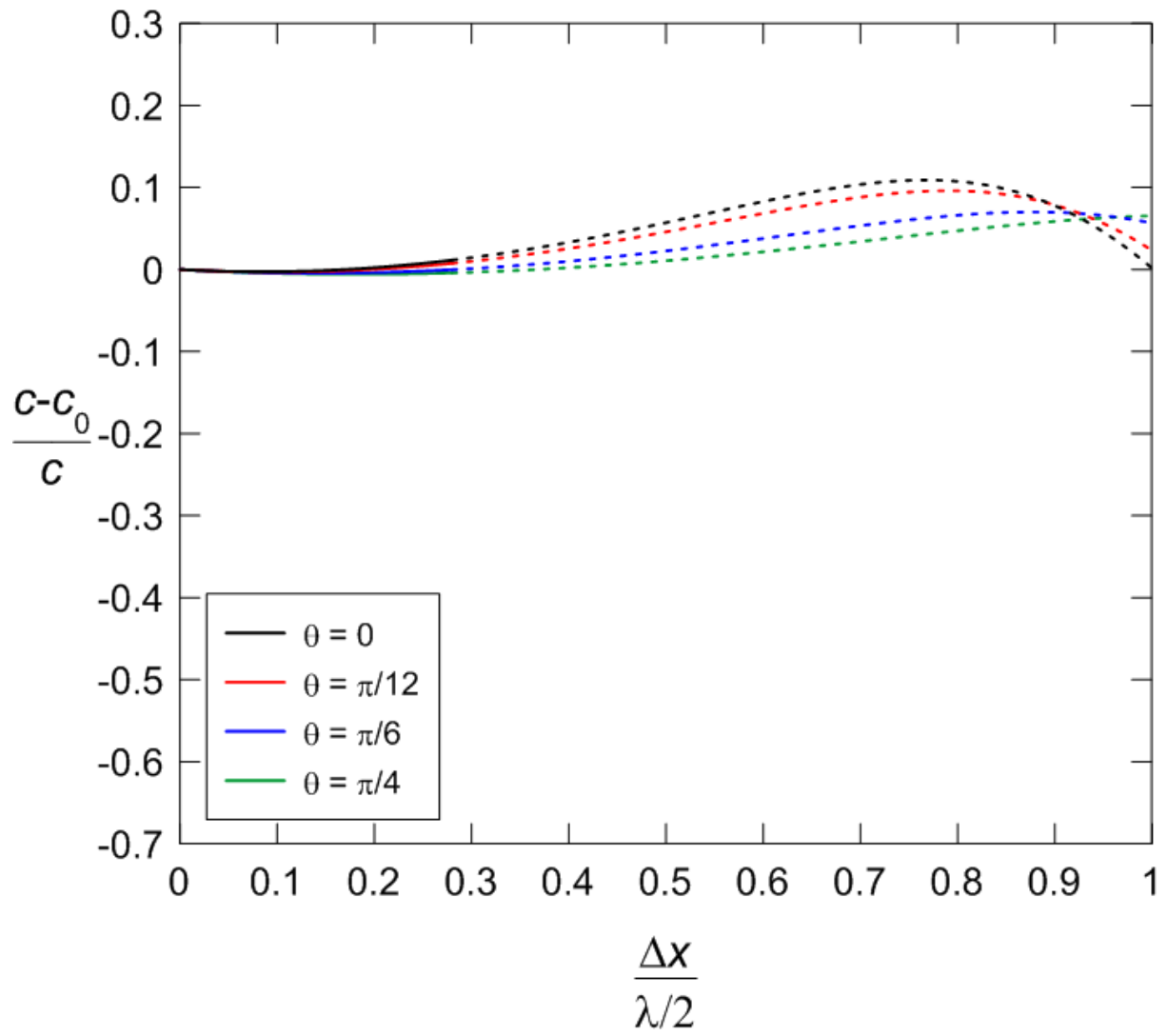


Fig. 16. Relative wave speed errors of the proposed method ($\alpha = 0.045$ and $\text{CFL} = 0.22$) for various propagation angles considering the consistent mass matrix (dotted lines: discarded wave modes).

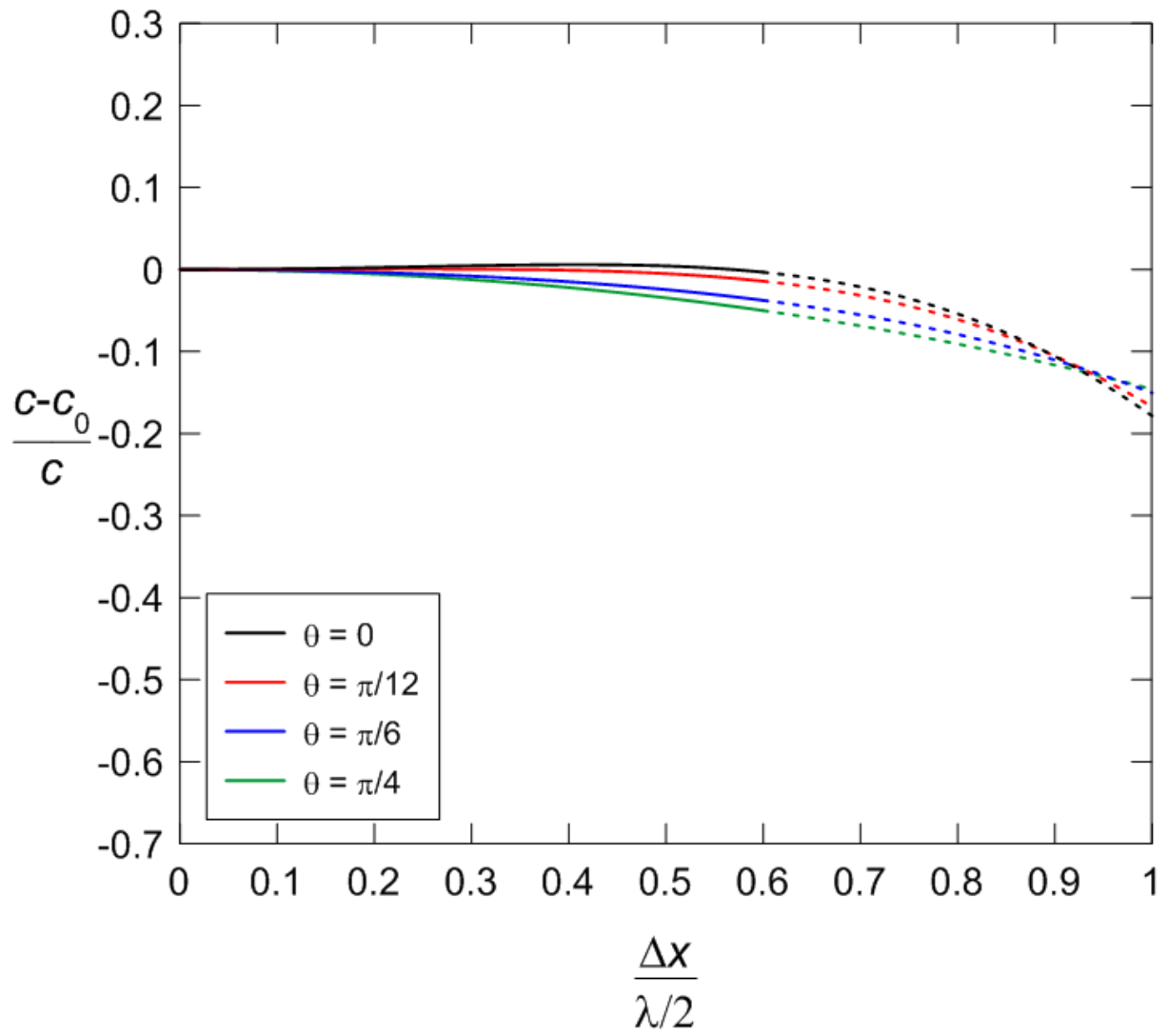


Fig. 17. Relative wave speed errors of the Bathe method (CFL = 1) for various propagation angles considering the consistent mass matrix (dotted lines: discarded wave modes).

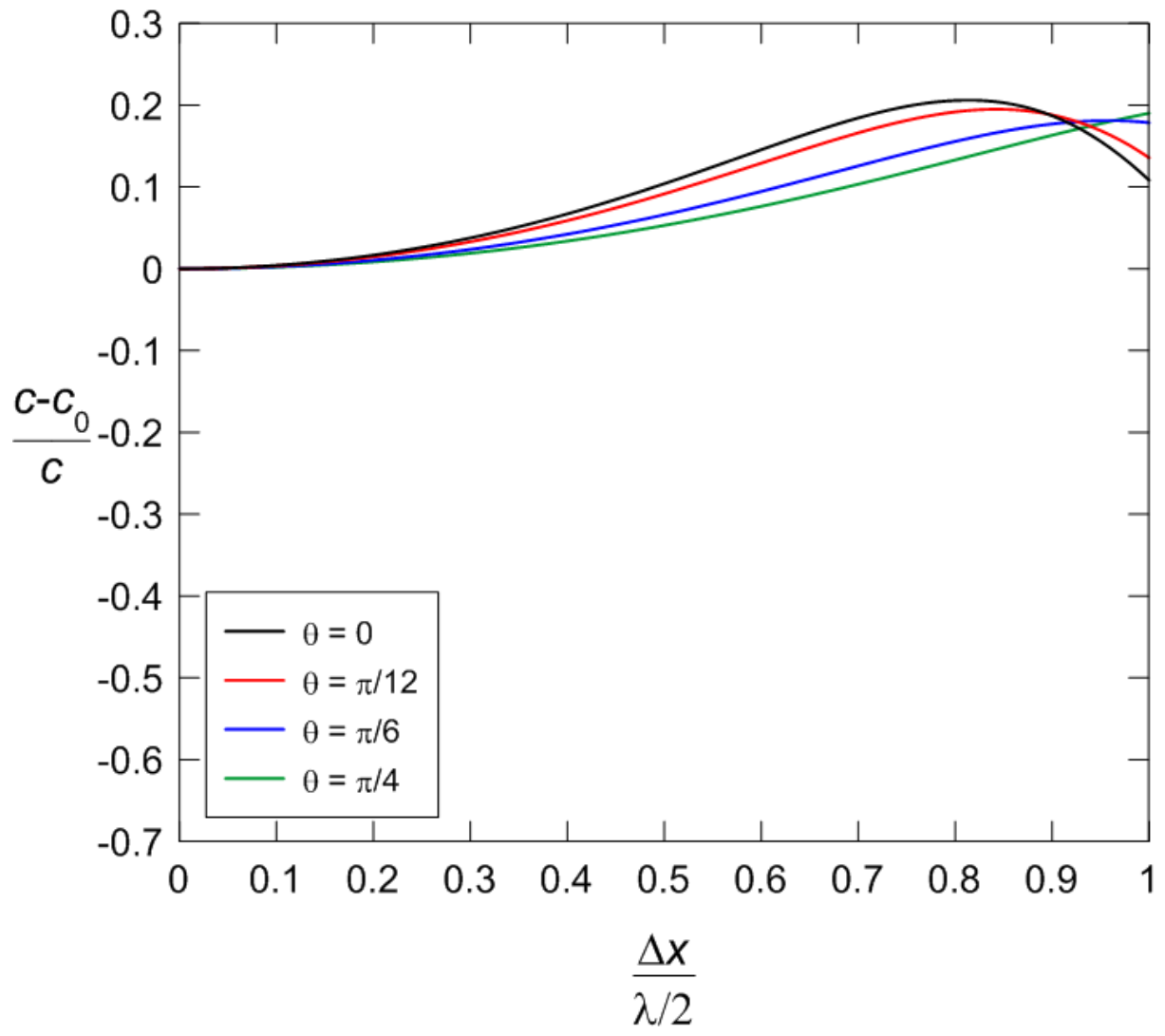


Fig. 18. Relative wave speed errors of the central difference method (CFL = 0.1) for various propagation angles considering the consistent mass matrix.

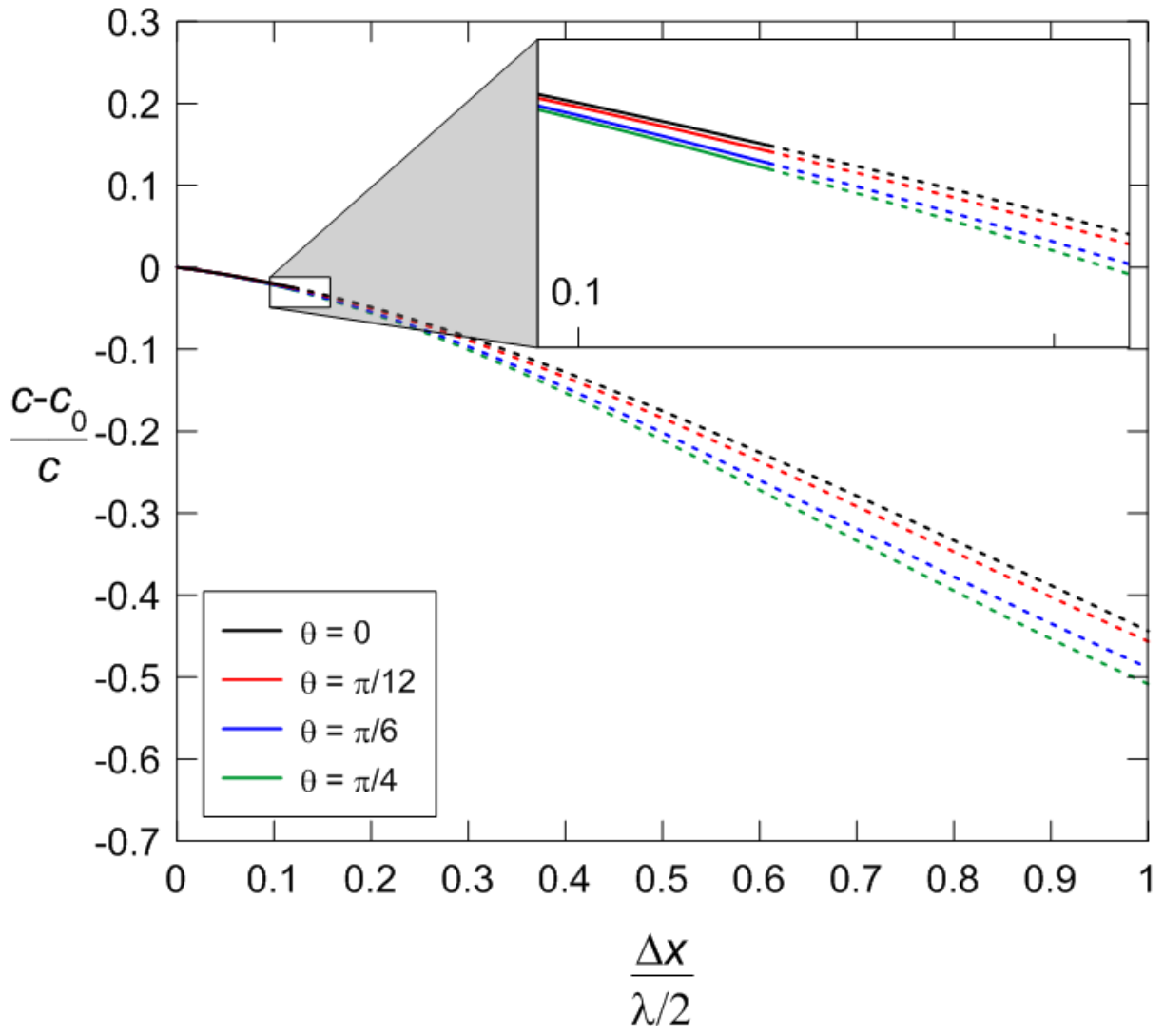


Fig. 19. Relative wave speed errors of the proposed method ($\alpha = 0.045$ and $\text{CFL} = 0.5$) for various propagation angles considering the lumped mass matrix (dotted lines: discarded wave modes).

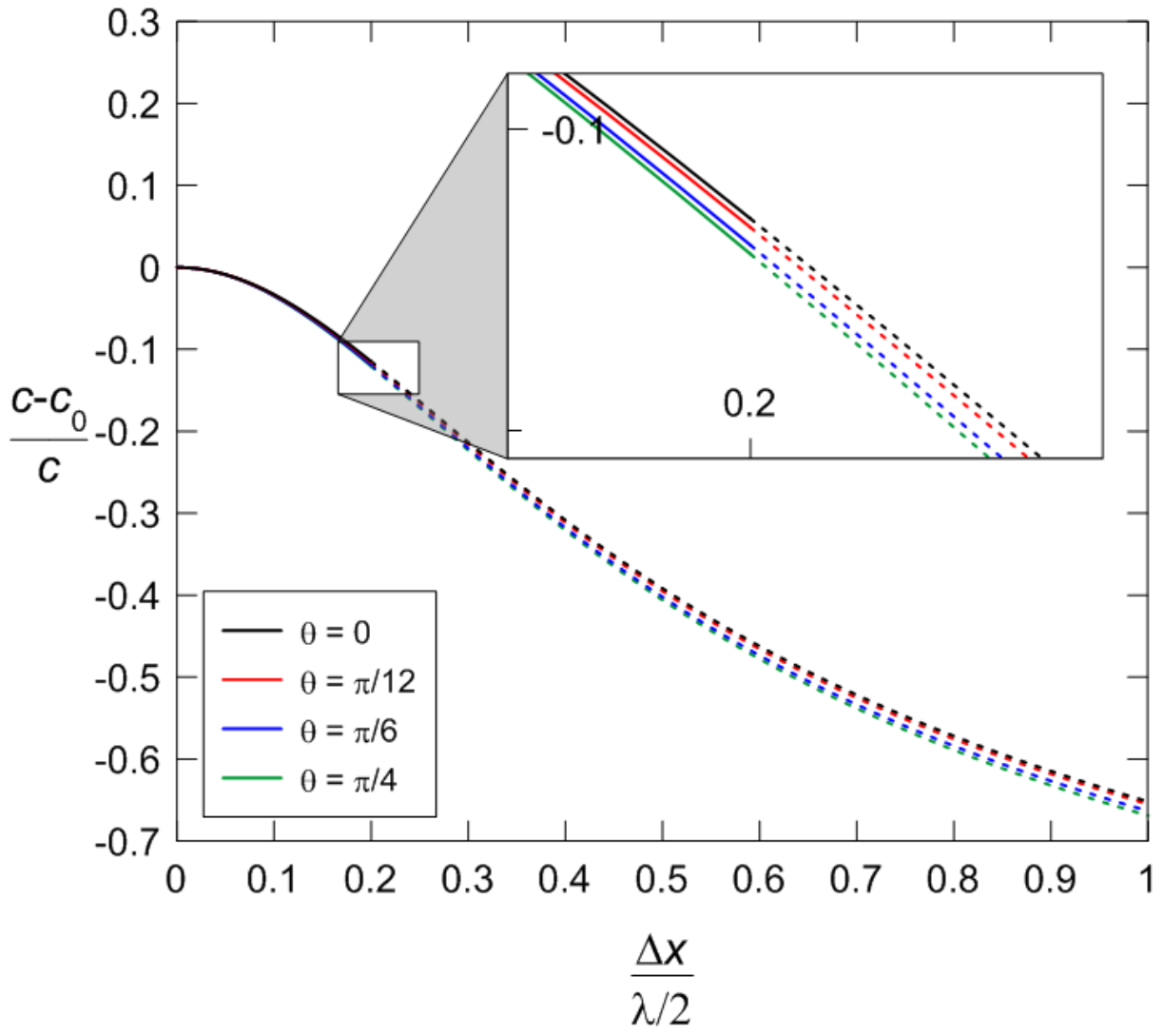


Fig. 20. Relative wave speed errors of the Bathe method (CFL = 3) for various propagation angles considering the lumped mass matrix (dotted lines: discarded wave modes).

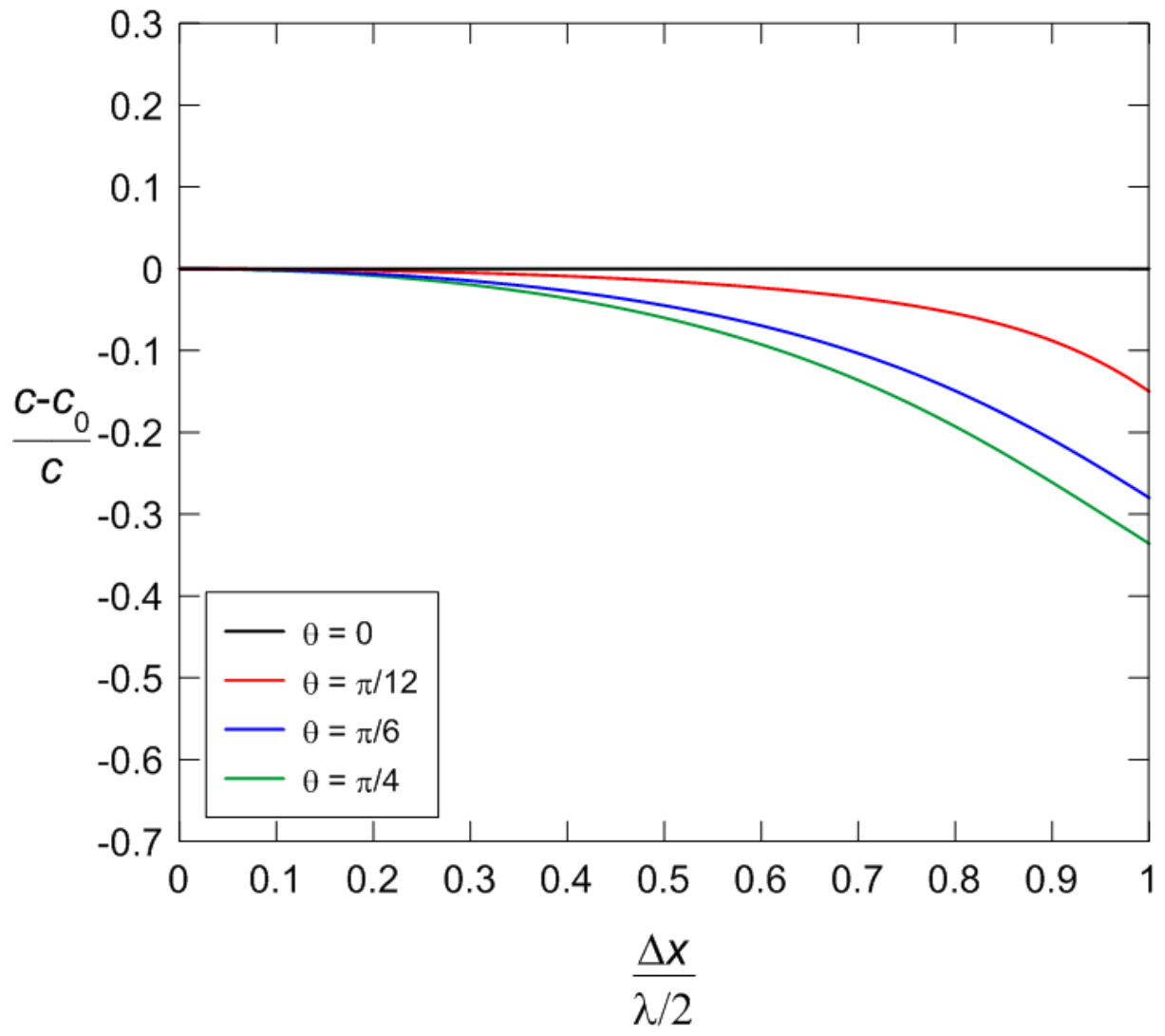


Fig. 21. Relative wave speed errors of the central difference method (CFL = 1) for various propagation angles considering the lumped mass matrix.

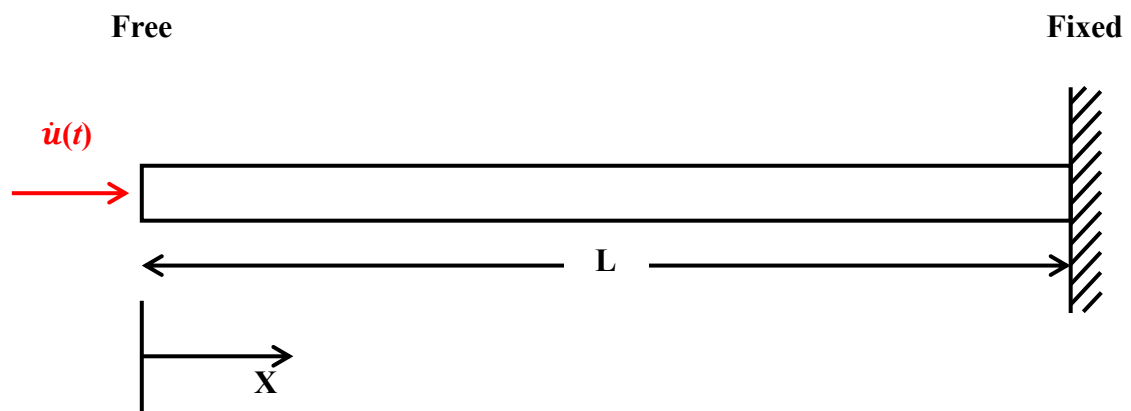


Fig. 22. 1D elastic bar impact problem: initial displacement, velocity, and acceleration are 0;
end conditions of the bar are free-fixed.

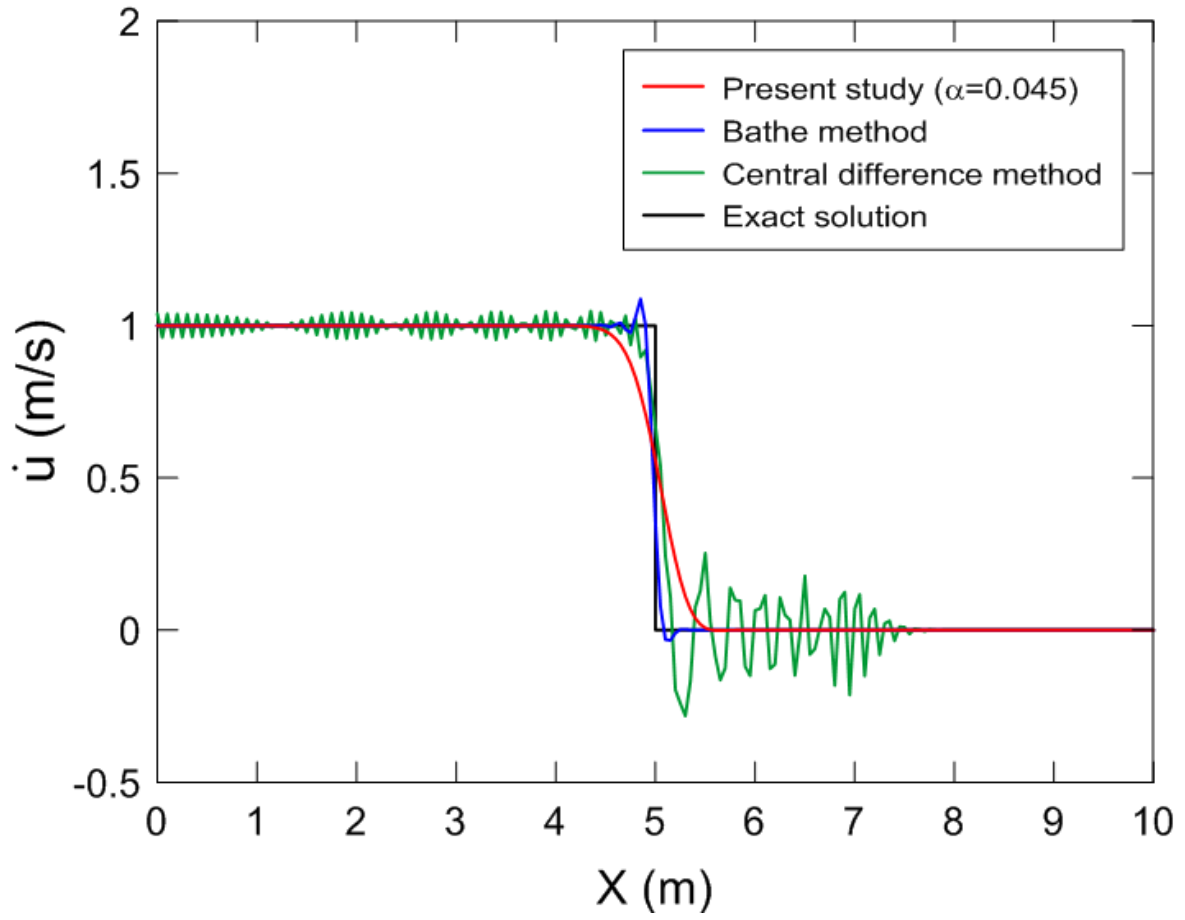


Fig. 23. Velocity distributions using the consistent mass matrix with various time integration schemes at $t = 1$ ms; CFL numbers for the proposed method, Bathe method, and central difference method are 0.22, 1, and 0.1, respectively.

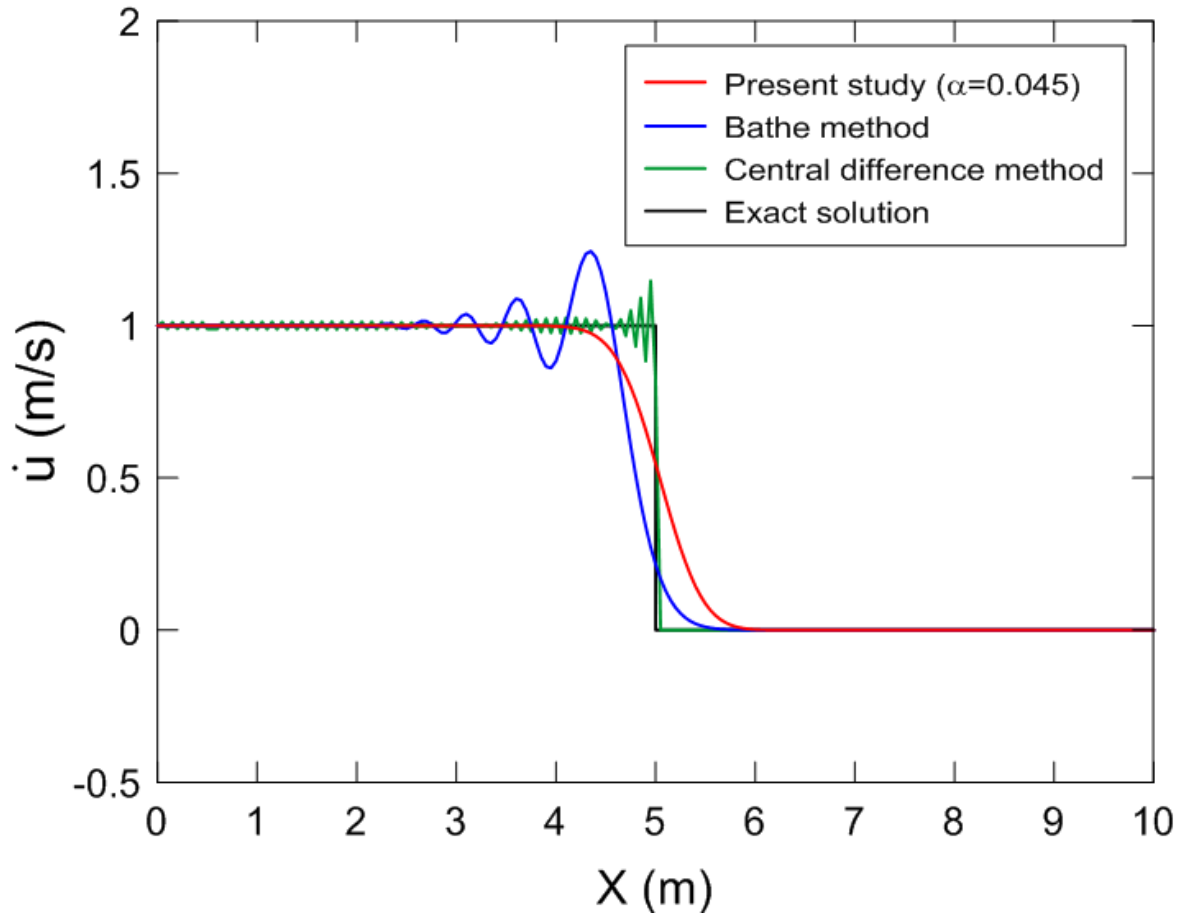


Fig. 24. Velocity distributions using the lumped mass matrix with various time integration schemes at $t = 1$ ms; CFL number for the proposed method, Bathe method, and central difference methods are 0.5, 3, and 1, respectively.

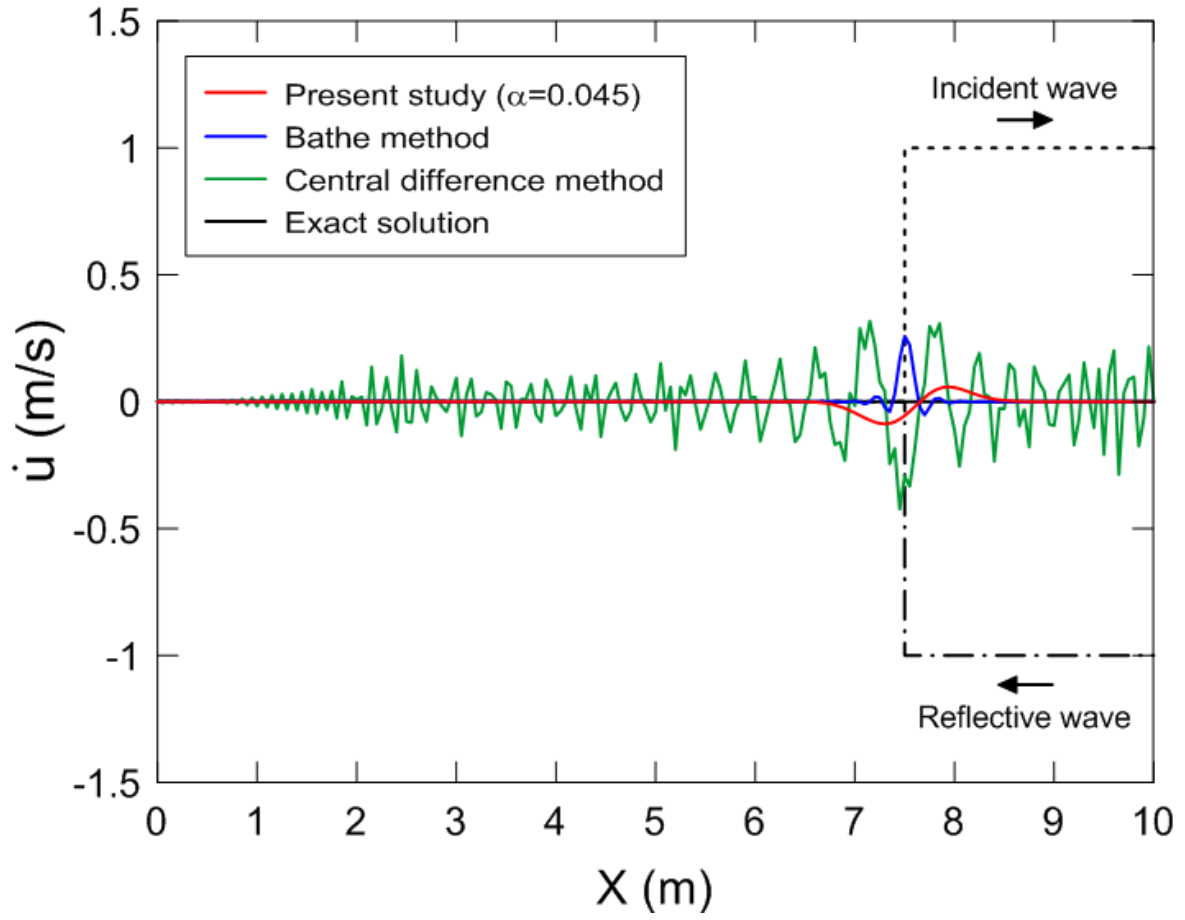


Fig. 25. Velocity distributions using the consistent mass matrix with various time integration schemes at $t = 2.5$ ms; CFL numbers for the proposed method, Bathe method, and central difference method are 0.22, 1, and 0.1, respectively.

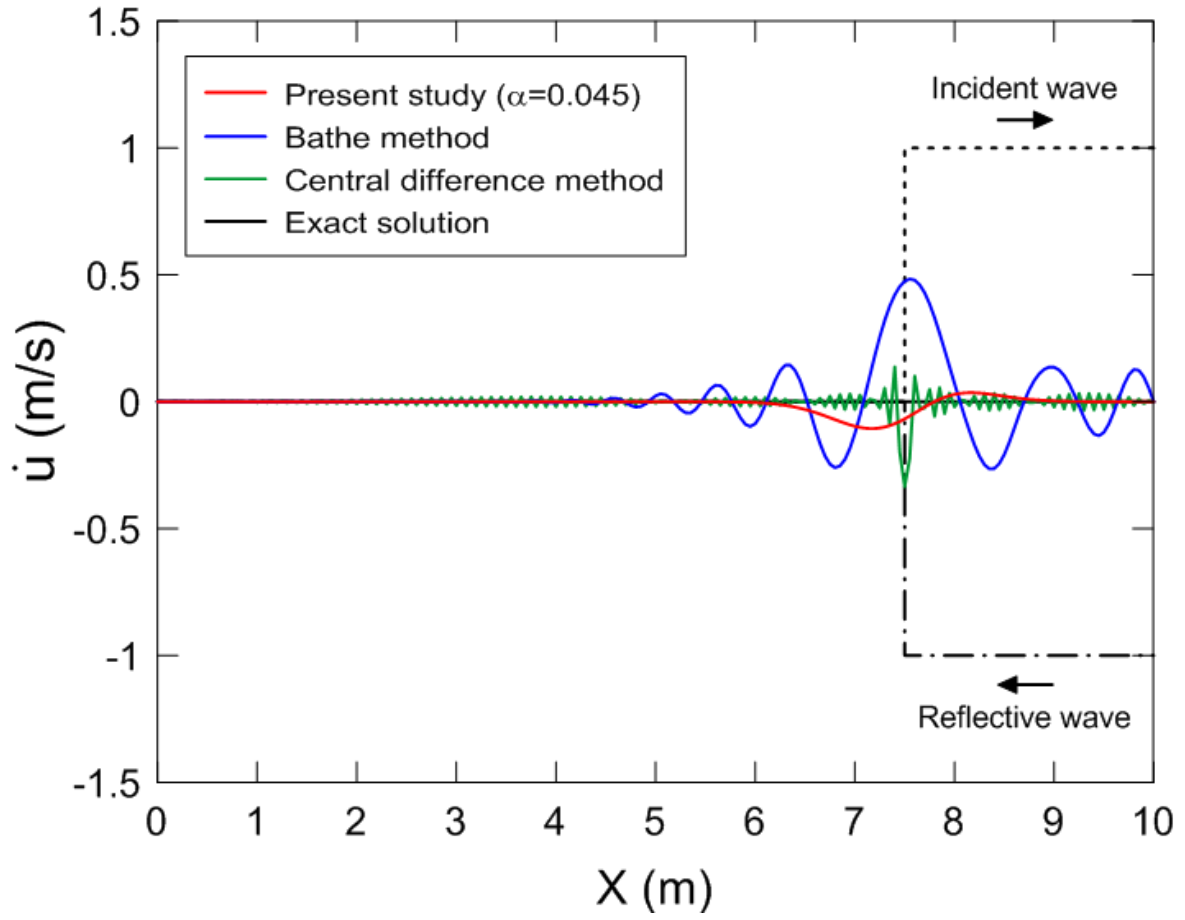


Fig. 26. Velocity distributions using the lumped mass matrix with various time integration schemes at $t = 2.5$ ms; CFL numbers for the proposed method, Bathe method, and central difference method are 0.5, 3, and 1, respectively.

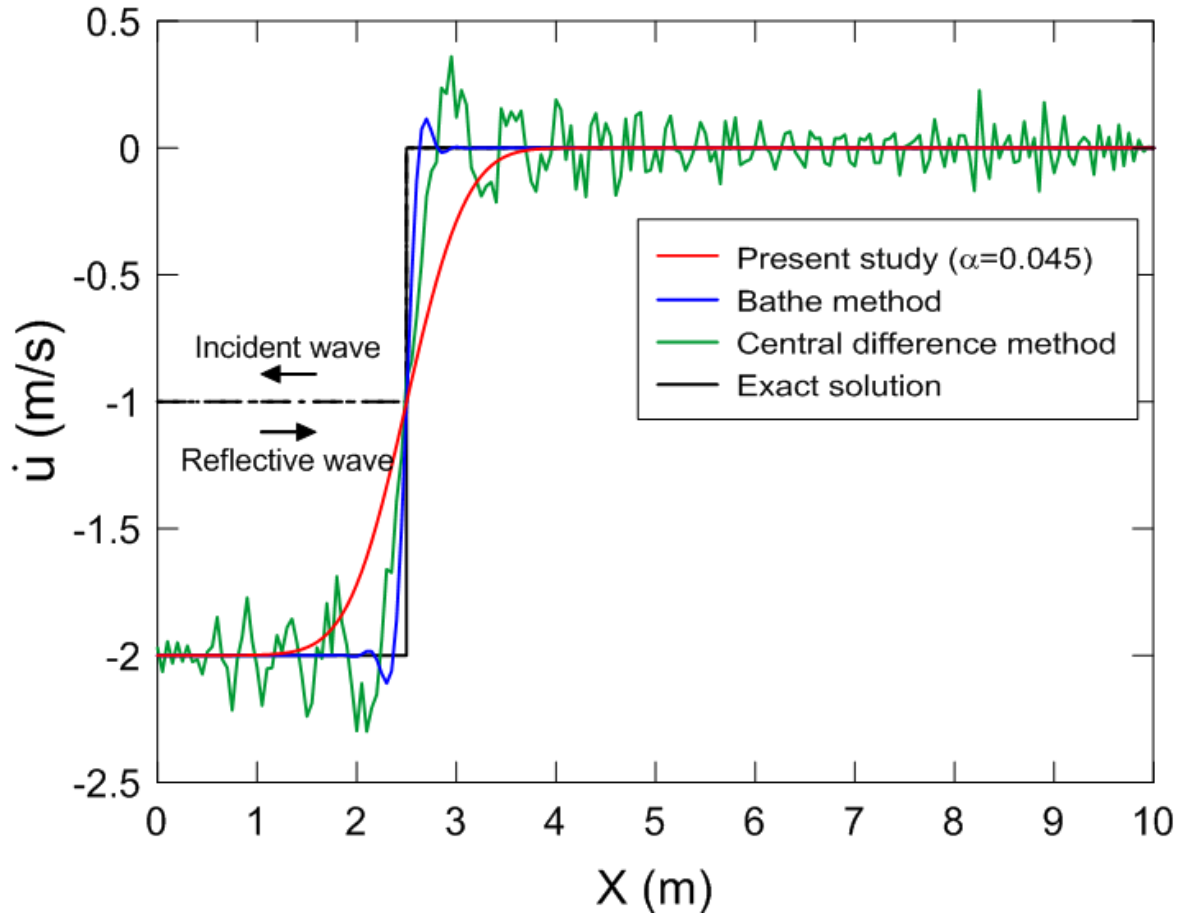


Fig. 27. Velocity distributions using the consistent mass matrix with various time integration schemes at $t = 4.5$ ms; CFL numbers for the proposed method, Bathe method, and central difference method are 0.22, 1, and 0.1, respectively.

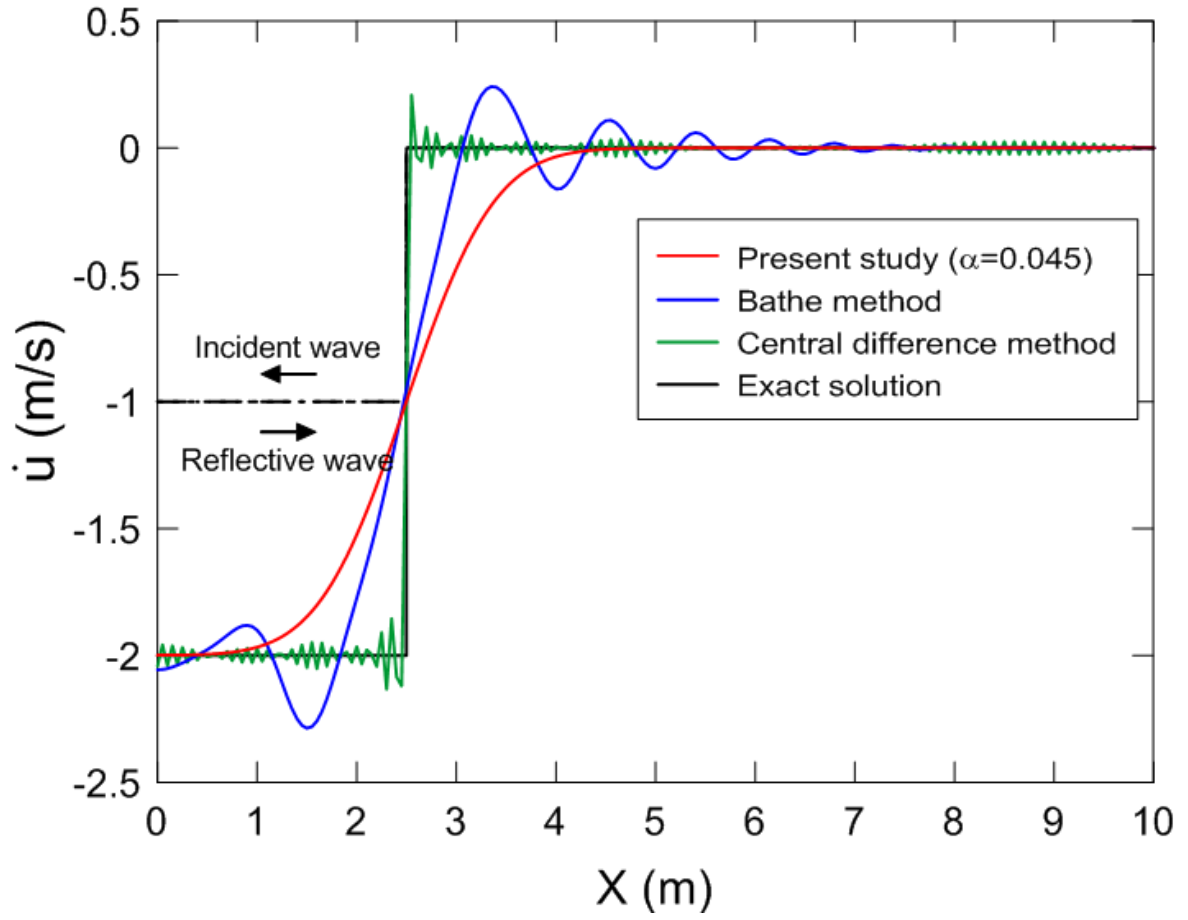


Fig. 28. Velocity distributions using the lumped mass matrix with various time integration schemes at $t = 4.5$ ms; CFL numbers for the proposed method, Bathe method, and central difference method are 0.5, 3, and 1, respectively.

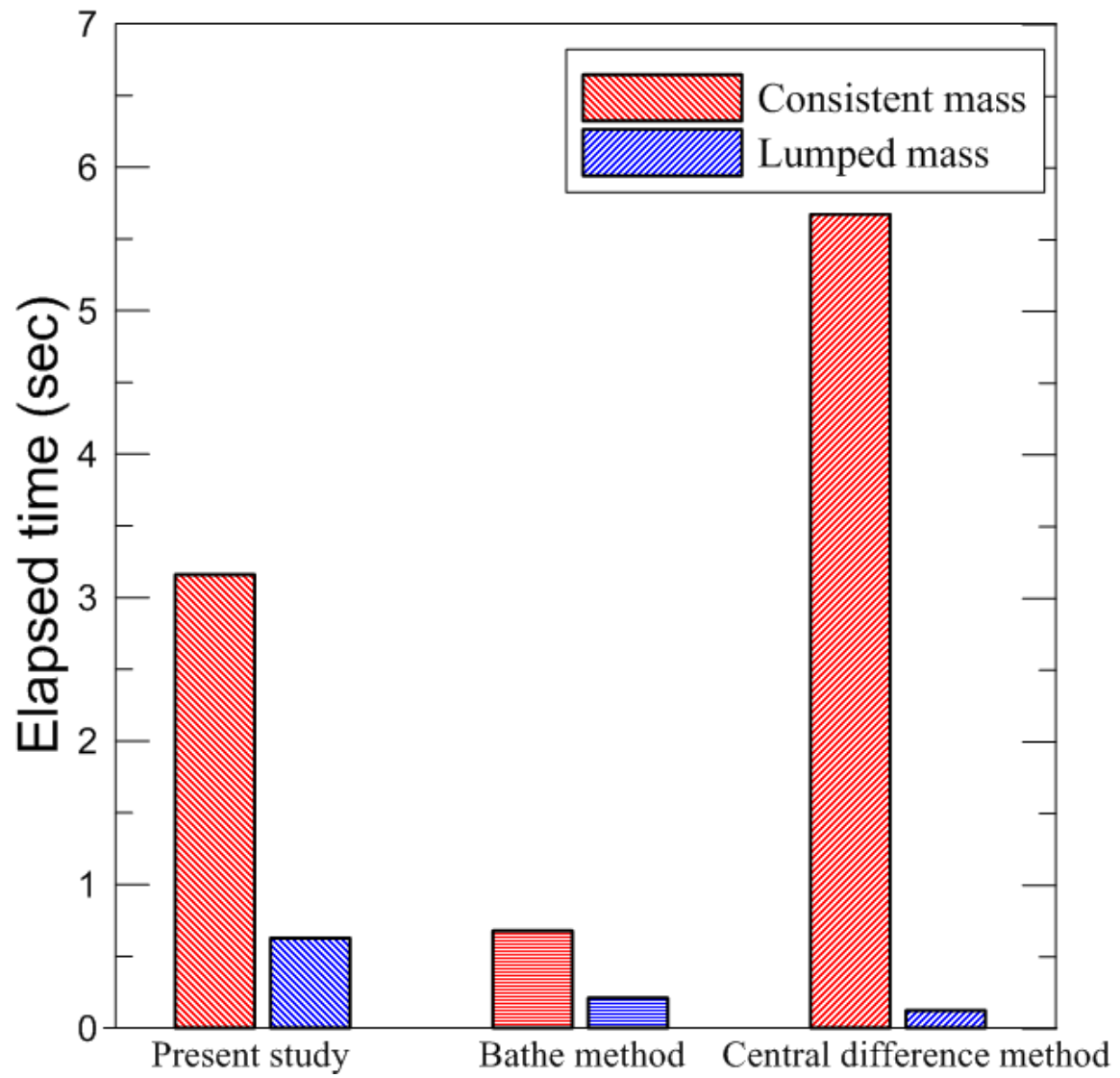


Fig. 29. Required computational efforts of various methods considering the 1D reflection problem at time $t = 4.5\text{ms}$, 200 element meshes.

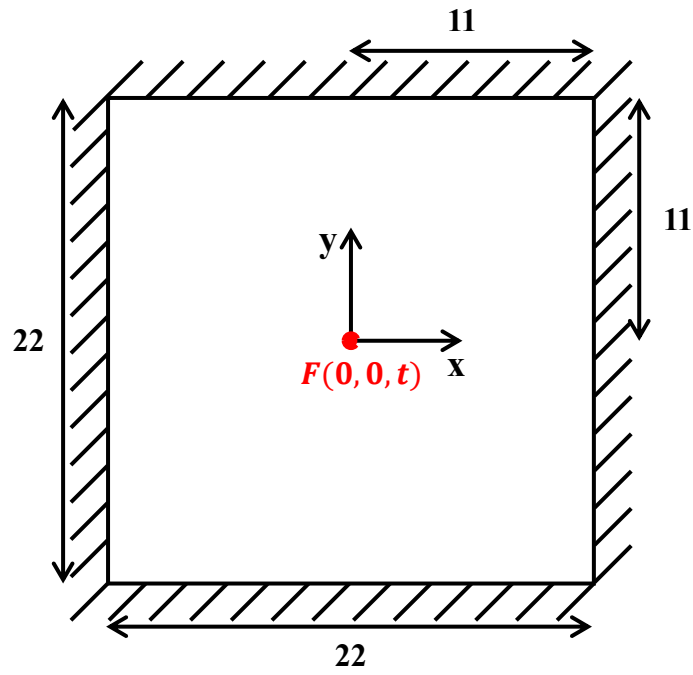
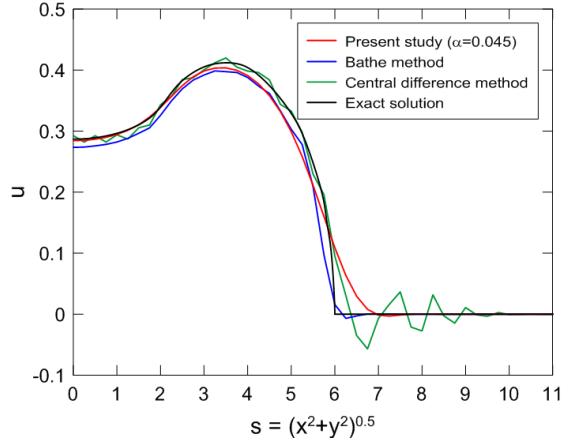
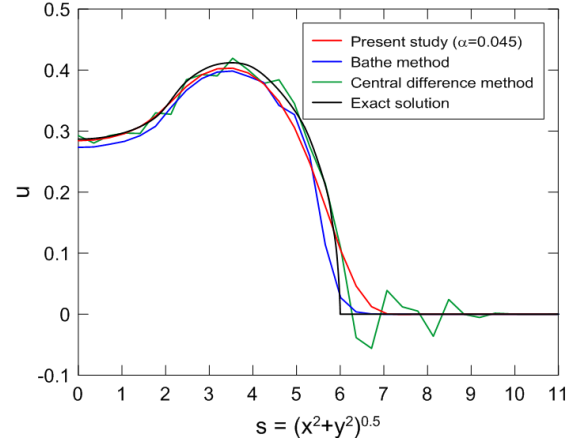


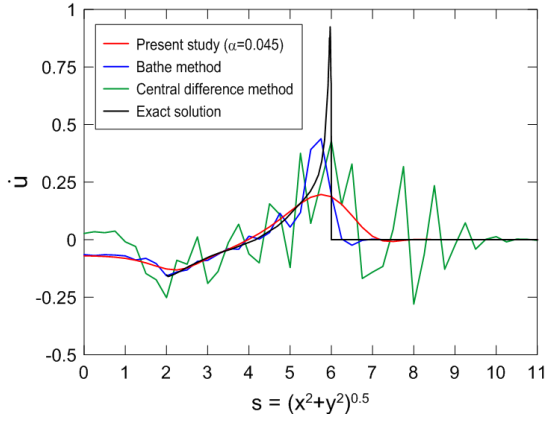
Fig. 30. Pre-stress membrane problem: initial displacement, velocity, and acceleration are 0 and exact wave speed is 1.0.



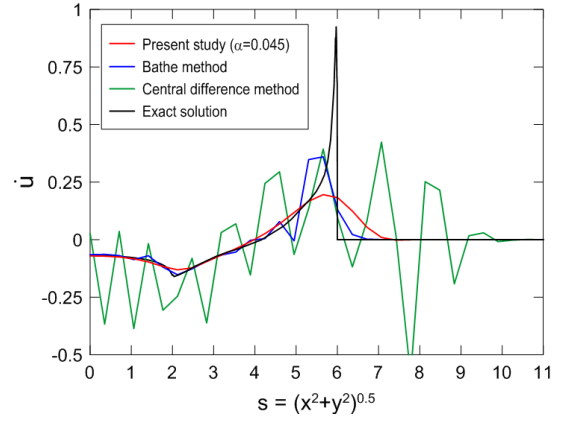
(a) Displacement when $\theta = 0$



(b) Displacement when $\theta = \frac{\pi}{4}$

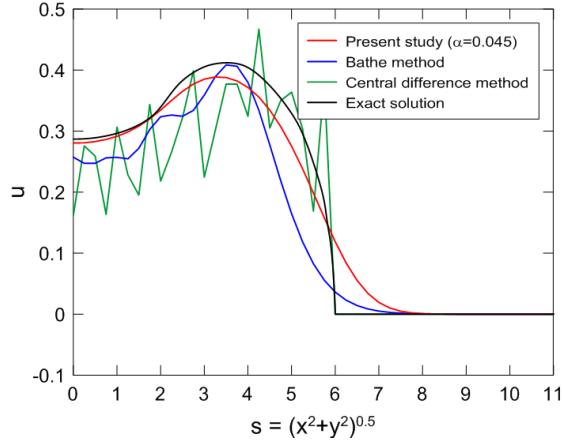


(c) Velocity when $\theta = 0$

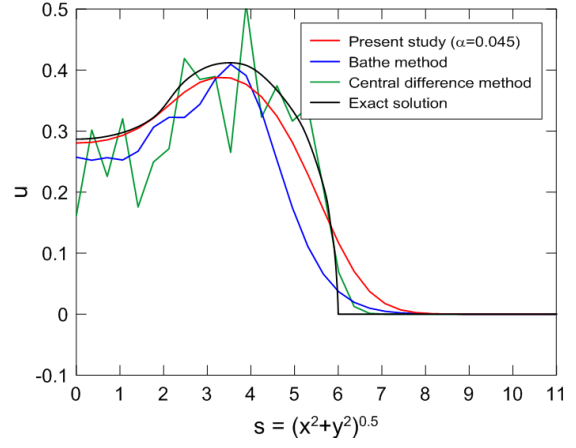


(d) Velocity when $\theta = \frac{\pi}{4}$

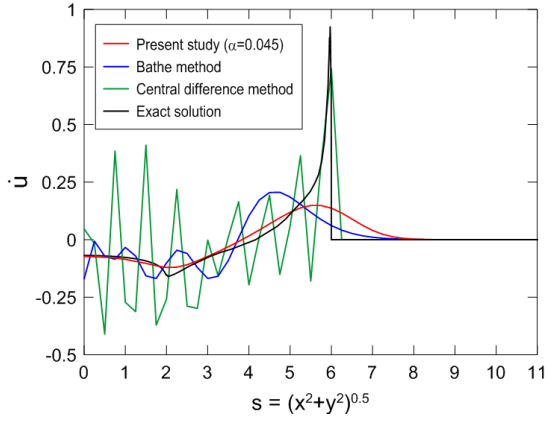
Fig. 31. Displacement and velocity variation of three methods along the various propagating angles for consistent mass, at time $t = 6$, 88×88 finite element mesh.



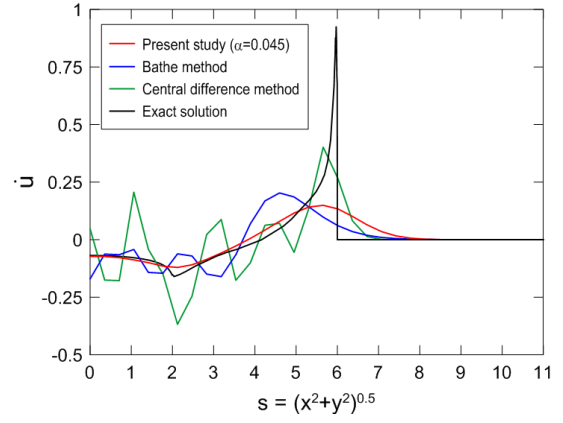
(a) Displacement when $\theta = 0$



(b) Displacement when $\theta = \frac{\pi}{4}$

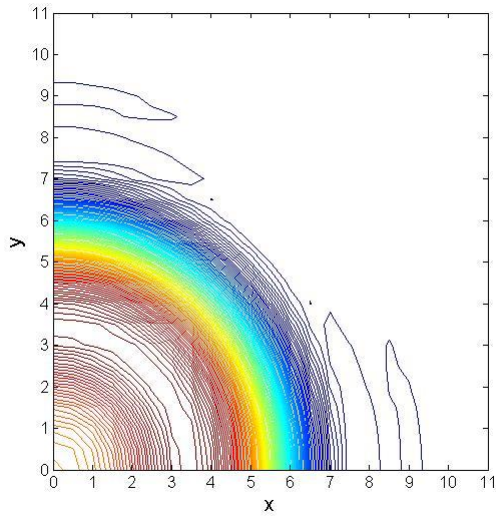


(c) Velocity when $\theta = 0$

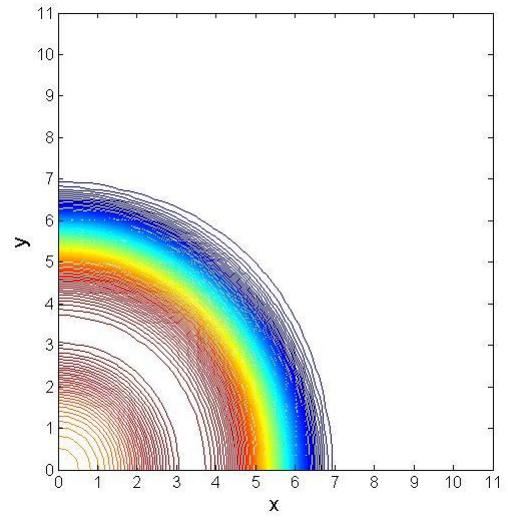


(d) Velocity when $\theta = \frac{\pi}{4}$

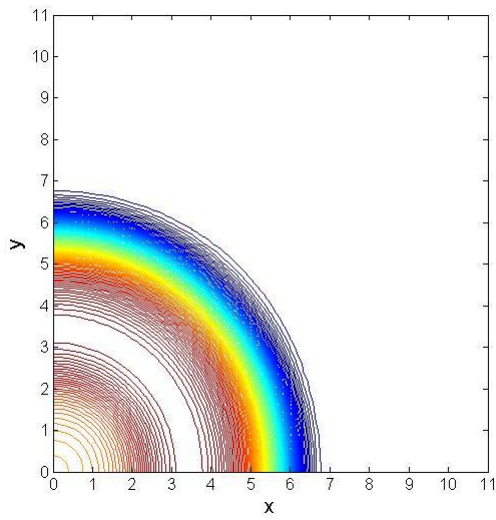
Fig. 32. Displacement and velocity variation of three methods along the various propagating angles for lumped mass, at time $t = 6, 88 \times 88$ finite element mesh.



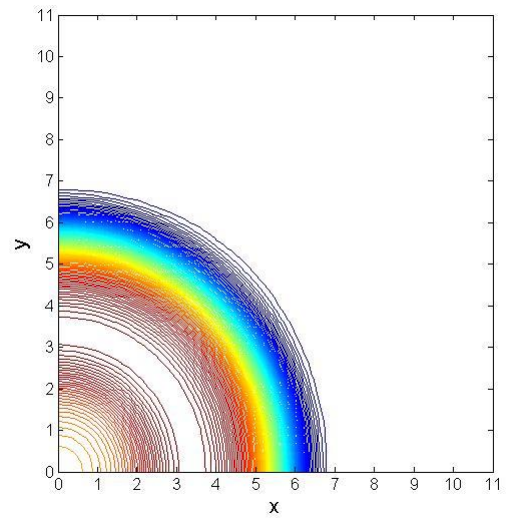
(a) 44 x 44



(b) 88 x 88

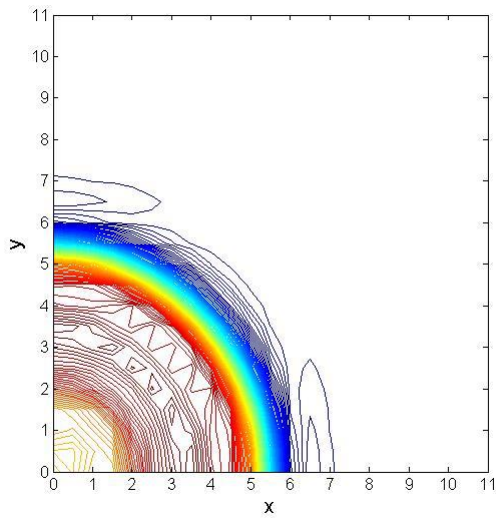


(c) 132 x 132

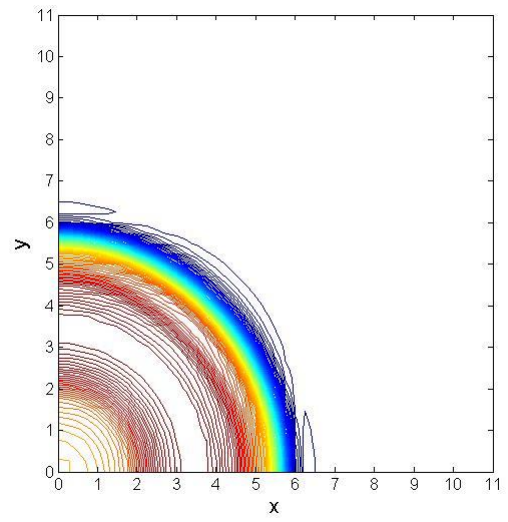


(d) 176 x 176

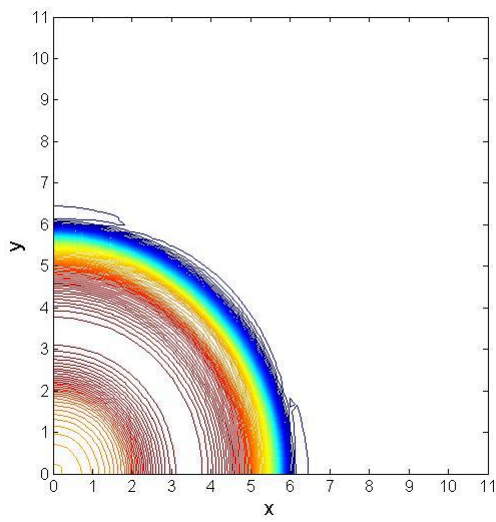
Fig. 33. Snapshots of displacement at $t = 6$ for the proposed method, when the consistent mass matrix considered.



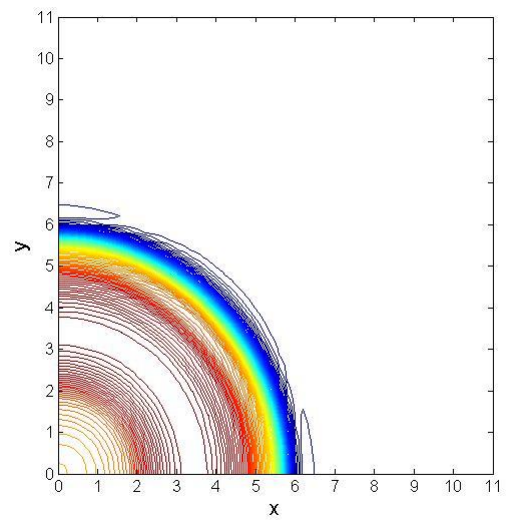
(a) 44 x 44



(b) 88 x 88

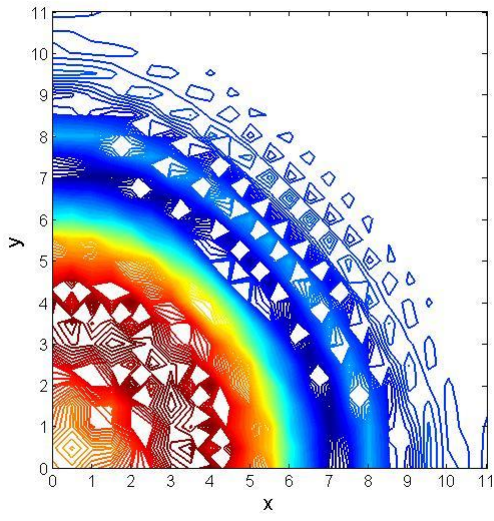


(c) 132 x 132

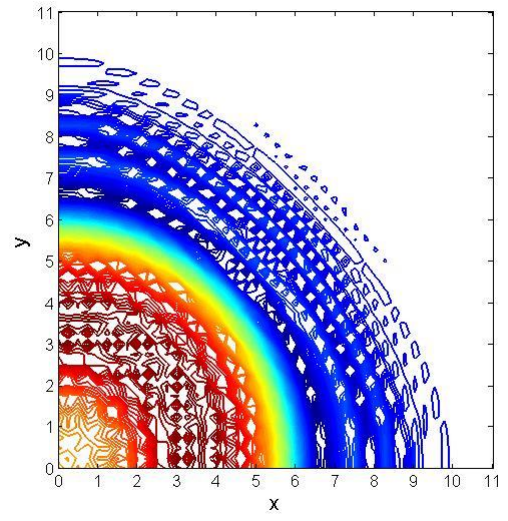


(d) 176 x 176

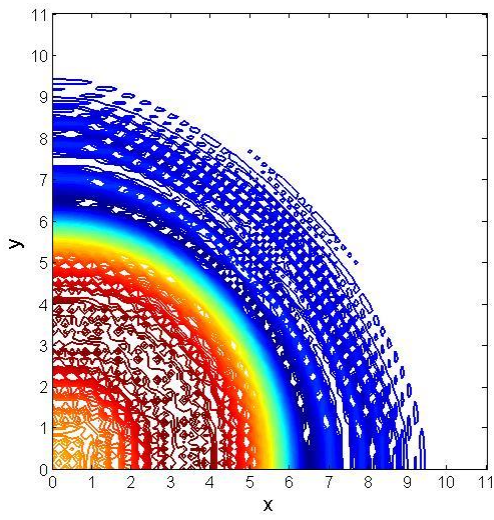
Fig. 34. Snapshots of displacement at $t = 6$ for the Bathe method, when the consistent mass matrix considered.



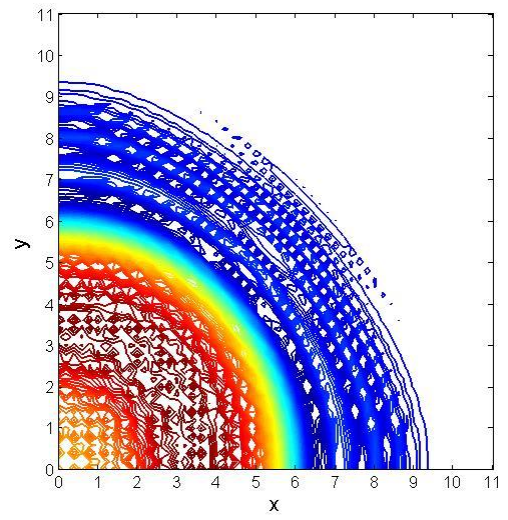
(a) 44×44



(b) 88×88

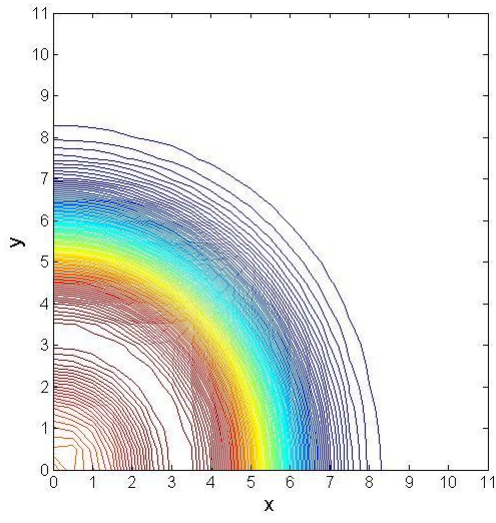


(c) 132×132

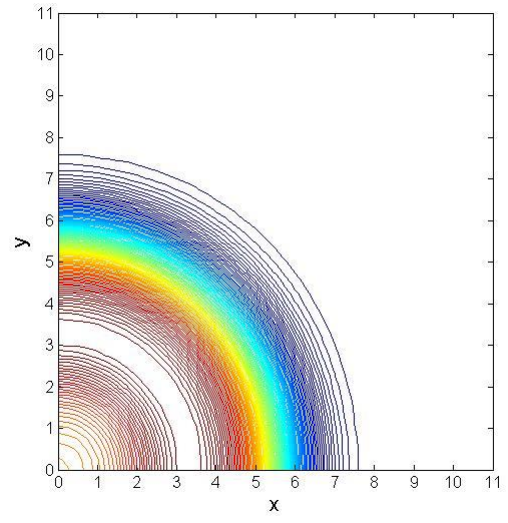


(d) 176×176

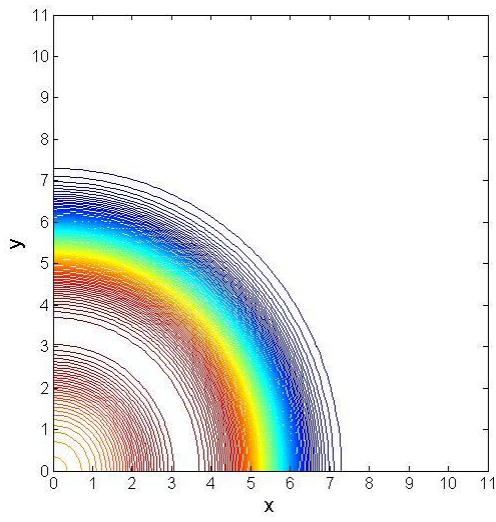
Fig. 35. Snapshots of displacement at $t = 6$ for the central difference method, when the consistent mass matrix considered.



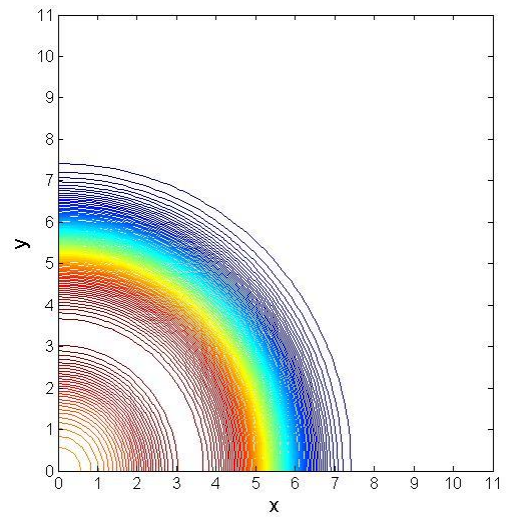
(a) 44 x 44



(b) 88 x 88

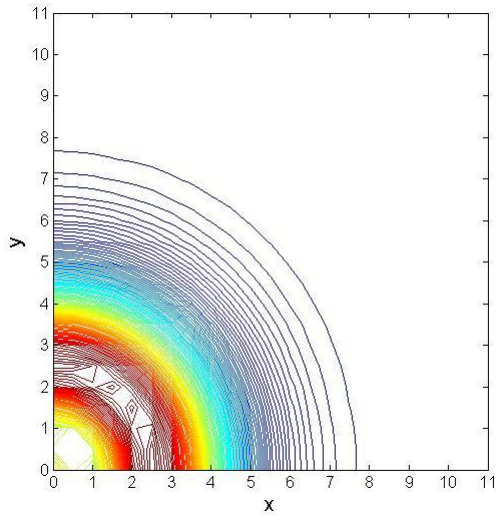


(c) 132 x 132

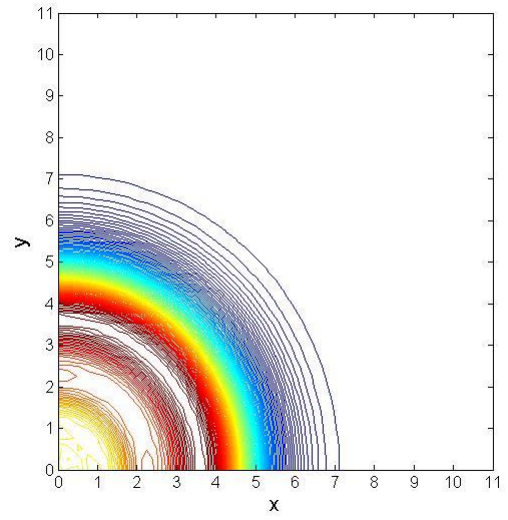


(d) 176 x 176

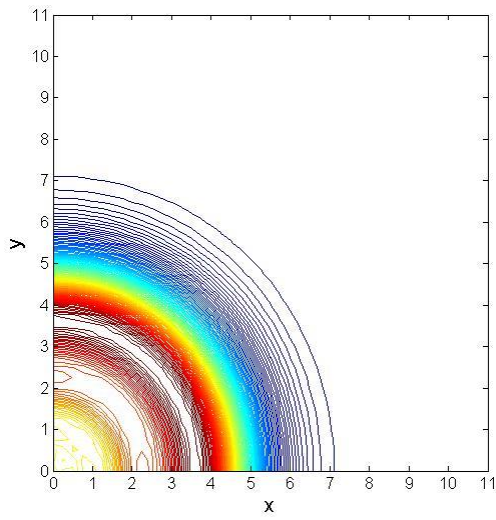
Fig. 36. Snapshots of displacement at $t = 6$ for the proposed method, when the lumped mass matrix considered.



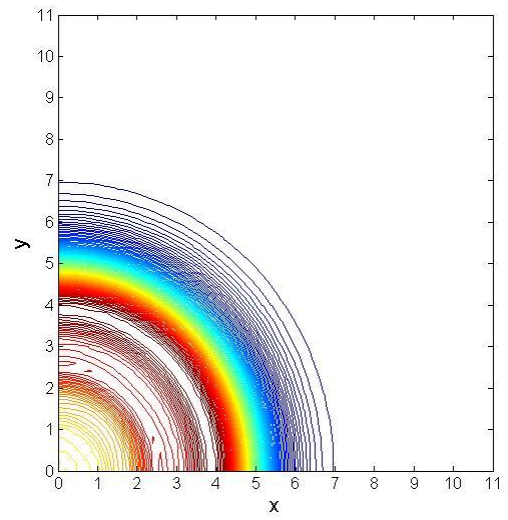
(a) 44 x 44



(b) 88 x 88

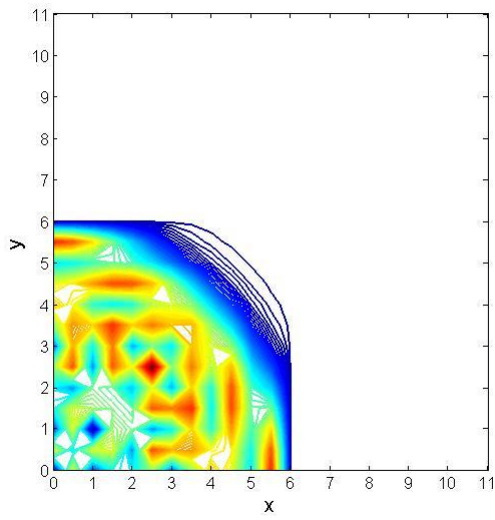


(c) 132 x 132

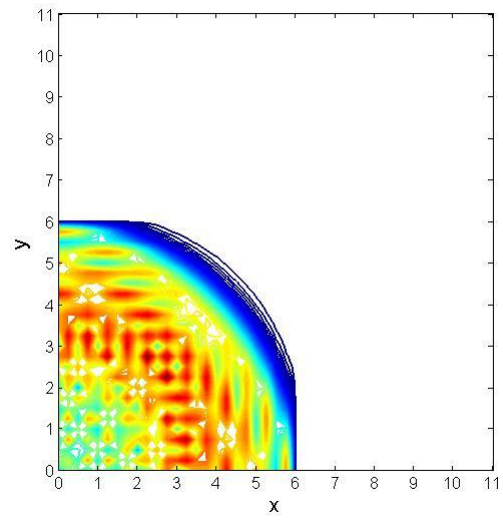


(d) 176 x 176

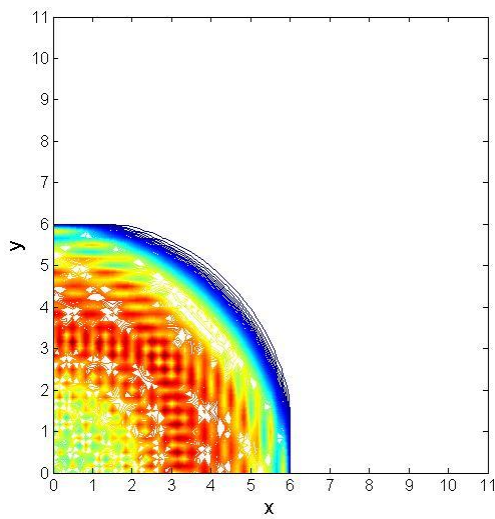
Fig. 37. Snapshots of displacement at $t = 6$ for the Bathe method, when the lumped mass matrix considered.



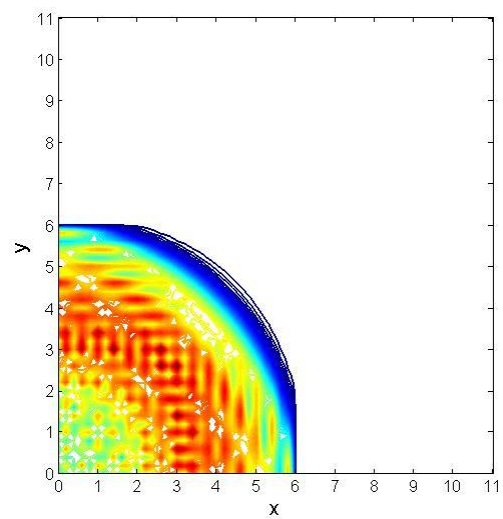
(a) 44×44



(b) 88×88



(c) 132×132



(d) 176×176

Fig. 38. Snapshots of displacement at $t = 6$ for the central difference method, when the lumped mass matrix considered.

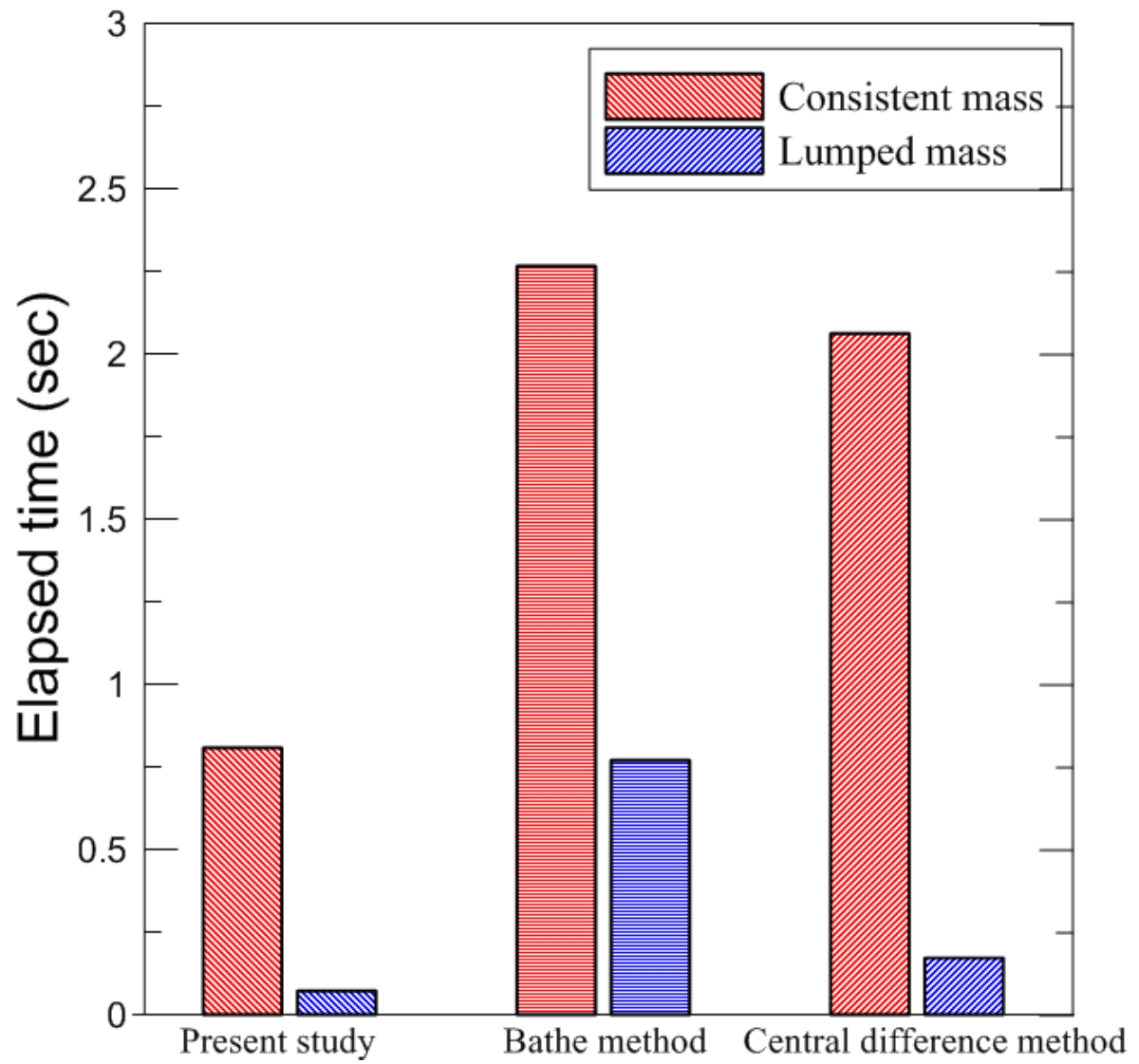


Fig. 39. Required computational efforts of various methods considering the 2D wave problem for 88 x 88 element mesh in section 3.2.